
Toward Culturally Aligned LLMs through Ontology-Guided Multi-Agent Reasoning

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Abstract

Large Language Models (LLMs) increasingly support culturally sensitive decision making, yet often exhibit misalignment due to skewed pre-training data and the absence of structured value representations. Existing methods can steer outputs, but often lack demographic grounding and treat values as independent, unstructured signals, reducing consistency and interpretability. We propose **OG-MAR**, an **O**ntology-**G**uided **M**ulti-**A**gent **R**easoning framework. **OG-MAR** summarizes respondent-specific values from the World Values Survey (WVS) and constructs a global cultural ontology by eliciting relations over a fixed taxonomy via competency questions. At inference time, it retrieves ontology-consistent relations and demographically similar profiles to instantiate multiple value-persona agents, whose outputs are synthesized by a judgment agent that enforces ontology consistency and demographic proximity. Experiments on regional social-survey benchmarks across four LLM backbones show that **OG-MAR** improves cultural alignment and robustness over competitive baselines, while producing more transparent reasoning traces.

1. Introduction

Large Language Models (LLMs) are predominantly trained on web-scale corpora that are unevenly distributed across regions and sociocultural contexts (Bender et al., 2021; Dodge et al., 2021; Achiam et al., 2023; Jiang et al., 2023; Touvron et al., 2023). As a result, they often inherit culture-default biases, prioritizing high-resource Western-centric viewpoints

while underrepresenting diverse cultural value systems (Durmus et al., 2023; Gallegos et al., 2024; Xie et al., 2024). These biases lead to systematic misalignment in culturally sensitive tasks, particularly ones involving social norms and value-based decisions (Karinshak et al., 2024; Pistilli et al., 2024; Tao et al., 2024). In response, several countries and organizations have developed localized LLMs to better reflect region-specific values (Zeng et al., 2022; Sengupta et al., 2023; Avramidis et al., 2024; Nguyen et al., 2024; Yoo et al., 2024). Nevertheless, cultural bias and value misalignment continue to pose challenges for LLM deployments (Hershcovich et al., 2022; Kreutzer et al., 2022).

Prior work has proposed several strategies to reduce cultural bias in LLMs. Role-assignment methods (Tao et al., 2024) steer behavior by specifying culturally grounded personas, while few-shot prompting uses curated cultural exemplars to guide generation (Choenni & Shutova, 2024). Retrieval-based approaches such as ValuesRAG (Seo et al., 2025) further ground outputs in external survey evidence to better match cultural preferences. More recently, multi-agent frameworks (Baltaji et al., 2024; Ki et al., 2025; Wan et al., 2025) simulate diverse viewpoints through agent interaction and deliberation. In particular, the debate-only framework (Ki et al., 2025) relies on iterative critique and refinement to improve cultural adaptability.

Despite their promise, existing approaches share several fundamental limitations: (1) they often depend on implicit cultural assumptions that are weakly grounded in empirical value distributions, making outputs brittle and sensitive to prompting choices; (2) even with external evidence, cultural values are frequently treated as independent signals, missing structural relationships and cross-topic dependencies; (3) aggregation and multi-agent methods can boost robustness and diversity but using multiple agents without concrete value structure or grounding often reduces interpretability, offering limited visibility into why specific viewpoints emerge.

To address these issues, we propose **OG-MAR**, an ontology-driven cultural reasoning framework that integrates structured value knowledge, demographic grounding, and multi-agent simulation. We specifically use the World Values Survey (WVS) (Zhao et al., 2024) as an empirically grounded

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retrieval corpus capturing diverse value distributions across regions. Raw survey responses are converted into topic-aware value summaries, and a global cultural ontology is built using expert-designed Competency Questions (CQs) (Gruninger, 1995; Grüninger & Fox, 1995), region-stratified LLM reasoning, and human-guided consolidation. At inference time, we retrieve ontology-consistent value structures and demographically similar individuals, instantiate value-persona agents for culturally grounded reasoning, and synthesize their outputs via a principled judgment mechanism.

We evaluate our framework on six regional benchmarks from major social surveys spanning East Asia, South Asia, Europe, North America, Latin America, and Africa. Results show consistent gains in cultural alignment, robustness across question types, and interpretability of reasoning traces. Quantitative analyses and qualitative case studies further indicate that ontology-guided multi-agent simulation offers a scalable and reliable path to culturally aligned LLM inference.

2. Related Work

2.1. Evaluation of LLM’s Cultural Alignment

Large Language Models (LLMs) exhibit strong linguistic ability, yet geographic and linguistic skew in pretraining data can embed dominant-region norms as implicit defaults, causing cultural misalignment and potential inequities across contexts (AlKhamissi et al., 2024). Prior work evaluated these biases using stereotype-focused benchmarks (Nadeem et al., 2021), open-ended generation measures (Dhamala et al., 2021), and task-level tests such as ambiguous question answering (Parrish et al., 2022). Later studies measured cultural alignment by comparing model outputs with representative value surveys like the World Values Survey (Haerpfer et al., 2022). Recent benchmarks further assess culture-specific everyday knowledge (Chiu et al., 2024), cross-national norm adaptation under different cultural frames (Rao et al., 2025), and value structure in open-ended generations through cultural psychology lenses (Karinshak et al., 2024). Building on this line of work, we use six regionally diverse survey datasets to evaluate value alignment across a broad set of cultural contexts.

2.2. Mitigating Cultural Bias in LLMs

Cultural bias mitigation for Large Language Models has evolved from in-context prompting to structured, evidence-grounded methods. Cultural prompting steers outputs by specifying a cultural frame (Tao et al., 2024), while Anthropological Prompting adds richer context and reasoning for underrepresented personas (AlKhamissi et al., 2024). To reduce sensitivity to examples and language, self-alignment selects culturally aligned demonstrations for in-context learn-

ing (Choenni & Shutova, 2024). When demonstrations are limited, ValuesRAG retrieves cultural and demographic cues as external evidence (Seo et al., 2025). Agentic approaches further improve reliability and parity through multi-agent debate and planning–critique–refinement pipelines (Ki et al., 2025; Wan et al., 2025). Despite this shift toward grounding and multi-agent reasoning, most methods still model cultural knowledge as unstructured, motivating ontology engineering to capture explicit value relationships.

2.3. Ontology Engineering with LLMs

An ontology is a formal specification of domain concepts and their relations (Gruber, 1993), supporting consistent and interpretable retrieval, integration, and reasoning. METHONTOLOGY (Fernández-López et al., 1997), On-To-Knowledge (Sure et al., 2004), and NeOn (Suárez-Figueroa et al., 2011) provide structured lifecycles and emphasize reuse. Recent work uses LLMs for ontology extraction via zero-shot prompting (Babaei Giglou et al., 2023) or fine-tuning (Mateiu & Groza, 2023), and increasingly automates end-to-end development: CQbyCQ matches novice-level performance (Saeedizade & Blomqvist, 2024), while Memoryless CQbyCQ and Ontogenia improve context efficiency and reasoning quality (Lippolis et al., 2025). Despite the benefits of ontology-based structuring for stereotype reduction, it remains underused in cultural-bias mitigation. We therefore propose a multi-agent framework for ontology-aware reasoning and pseudo-answer simulation.

3. Proposed Framework

We propose **OG-MAR**, **Ontology-Guided Multi-Agent Reasoning** framework that (i) summarizes respondent values under a fixed taxonomy, (ii) constructs CQ-derived cross-category relations as an ontology, (iii) retrieves ontology triples and demographically similar profiles for a query, and (iv) performs multi-persona simulation with ontology-constrained final adjudication.

3.1. Data Preprocessing & Structuring

3.1.1. TOPIC-AWARE VALUE SUMMARY GENERATION

Large-scale surveys capture cultural values through diverse question types, varying response scales, and heterogeneous answer formats. Directly operating on raw survey answers risks conflating unrelated signals and amplifying noise. To address this issue, we generate structured value summaries aligned with a predefined ontology taxonomy¹. Each World Values Survey (WVS) respondent record is decomposed into (1) demographic attributes and (2) values-related responses. Given the fixed ontology class set C ($|C| = 76$),

¹Taxonomy process and results are provided in Appendix H.

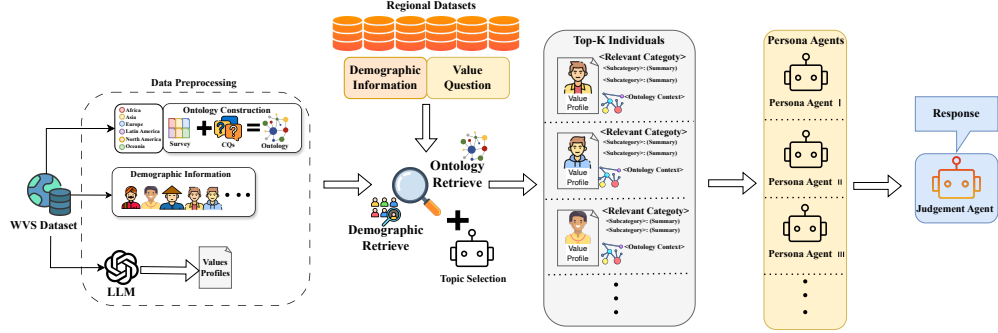


Figure 1. **Overall architecture of the OG-MAR framework.** The pipeline illustrates the overall architecture of **OG-MAR**. It begins with Data Preprocessing & Ontology Construction (left). During inference, for a given query and target demographics, it performs Ontology & Demographic Retrieval (center) to gather relevant context. This context is used to instantiate multiple Persona Agents (top right) whose outputs are synthesized by a Judgement Agent (bottom right) to produce the final, culturally aligned prediction.

which comprises 12 top-classes $c_1, \dots, c_{12}$ and their 64 sub-classes $c_{i,j}$ ², a Summarization Agent G_{sum} generates a concise, category-specific synopsis of the respondent’s stance within the semantic scope of each class in the taxonomy. The agent is instructed to summarize only information relevant to the given class. Formally, let $\mathcal{R}_i$ denote the raw response set for individual i . For each class $c \in C$, we obtain a category-conditioned synopsis:

$$s_i(c) = G_{\text{sum}}(\mathcal{R}_i \mid c). \quad (1)$$

Aggregating over all classes yields a structured value profile:

$$V_i = \{s_i(c) \mid c \in C\}. \quad (2)$$

Consequently, each individual is represented by a structured value profile that supports subsequent demographic grounding and persona simulation.

3.1.2. CQ-GUIDED ONTOLOGY RELATION CONSTRUCTION

To model relationships between value categories, we adopt a human-guided ontology construction process based on Competency Questions (CQs)³. Domain experts curate CQs, each designed to probe meaningful interactions between sub-classes of two given top-classes in the fixed taxonomy. For each CQ, we prompt a Large Language Model to describe sub-class-level relationships between the given top-classes. The model is constrained to: (i) use only the predefined taxonomy classes, (ii) avoid introducing any new classes, and (iii) focus solely on articulating relationships between sub-classes of the given top-classes.

Ontology Triple generation under cultural conditioning. To incorporate diverse cultural perspectives during ontol-

²We write c_i for the i -th top-class and $c_{i,j}$ for its j -th sub-class. The full taxonomy is detailed in Table 25.

³Details of the ontology construction process and CQ examples are provided in Appendix H, Table 24.

ogy construction, we condition the LLM on value profiles sampled from 120 individuals (20 per region) spanning six major world regions. Each CQ yields candidate relational statements, represented as ordered sentence triples:⁴

$$t_{a,b} = (c_a, p_{a,b}, c_b), \quad (3)$$

where c_a and c_b are short noun phrases describing sub-classes from the two queried top-classes, and $p_{a,b}$ is a natural-language relation verb phrase (distinct from p in top- p selection). Although these correspond to ontology classes and object properties, we express them as natural language sentences to maintain human interpretability and ensure alignment with the phrasing of the Competency Questions. We use parentheses to emphasize an ordered ontology triple of text rather than symbolic identifiers.

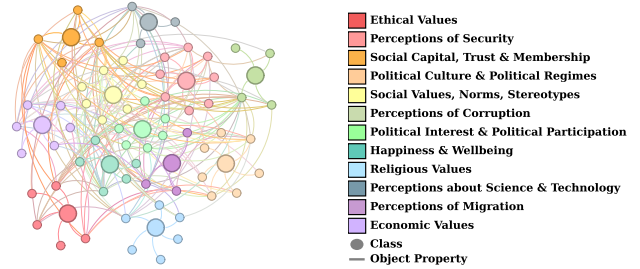


Figure 2. **Visualization of the final ontology structure.** The ontology consists of 12 top-classes and 64 sub-classes, connected by 150 object-property relations to form a comprehensive semantic network.

Consolidation and human review. Compared to traditional ontology engineering, the taxonomy remains fixed: no classes are merged, split, or added. Human experts review candidate object properties by (1) validating cultural plausibility, (2) editing relation descriptions for clarity and

⁴Construction prompts are provided in Appendix F, Table 16.

consistency, and (3) removing spurious or inconsistent relations⁵. The resulting ontology consists of a two-level class hierarchy. The 12 top-classes are $c_1, \dots, c_{12}$, and we write $c_{i,j}$ for the j -th sub-class under top-class c_i (with n_i sub-classes each). The complete taxonomy is

$$C = \{c_i\}_{i=1}^{12} \cup \{c_{i,j} \mid 1 \leq i \leq 12, 1 \leq j \leq n_i\}, \quad (4)$$

$$\sum_{i=1}^{12} n_i = 64, |C| = 76,$$

and a curated ontology triple set

$$T = \{t_h\}_{h=1}^{|T|}, \quad t_h = (c_a, p_{a,b}, c_b). \quad (5)$$

Here P denotes the set of relation texts, and we index the curated ontology triples as $\{t_h\}$, where each t_h corresponds to an ontology triple of the form $t_{a,b}$ above. Thus $T \subseteq \{c_{i,j}\} \times P \times \{c_{i,j}\}$ is the curated subset of CQ-derived cross-class relations, and $|T|$ is determined by expert review rather than by the combinatorics of $\{c_{i,j}\}$.

3.2. OG-MAR Inference Pipeline

3.2.1. QUERY ANALYSIS & CONTEXT RETRIEVAL

Given an input query q and target respondent demographics d_q , we retrieve (i) ontology-consistent triples and (ii) demographically grounded respondent profiles, which jointly define the context for downstream multi-agent simulation. We use D_q for the top- k top-classes, O_q for retrieved ontology triples, R_q for retrieved respondents, and V_q for their value profiles. Here, k , M , and K are the corresponding selection sizes.

(a) Topic/category identification. A Topic-Selection Model G_{topic} identifies relevant top-classes. We implement G_{topic} as a pretrained text encoder fine-tuned on WVS data labeled with the 12 top-classes. These selected top-classes determine the scope of sub-classes in subsequent ontology retrieval. Let $D = \{c_1, \dots, c_{12}\}$ denote the set of top-classes. The encoder outputs a logit score ℓ_u for each c_u , and the top- k top-classes are selected:

$$D_q = \{c_u \mid \ell_u \text{ is among the top-}k \text{ scores of } \{\ell_1, \dots, \ell_{12}\}\}. \quad (6)$$

(b) Ontology triple retrieval. We retrieve ontology knowledge in the form of triples, where each triple $t_h = (c_a, p_{a,b}, c_b)$ is treated as a single semantic unit. In this step, we *only* use node-level similarity within the sub-classes of the selected domains D_q . Specifically, for each sub-class $c \in \{c_{k,j} \mid c_k \in D_q\}$, we compute a node relevance:

$$\alpha(c) = \text{sim}(\mathbf{e}_q, \mathbf{e}_c), \quad (7)$$

⁵The collaborative construction process is detailed in Appendix H.1, and inter-group reliability in Appendix H.1.3.

where $\mathbf{e}_c$ is the embedding of the category text c .

We then score each ontology triple by the relevance of its endpoint nodes:

$$\alpha_{\text{triple}}(t_h) = \max(\alpha(c_a), \alpha(c_b)). \quad (8)$$

Finally, we restrict retrieval to ontology triples whose endpoints are both sub-classes belonging to top-classes in D_q , and select the top- M ontology triples by α_{triple} to form the ontology context O_q .

(c) Similar individual retrieval with dense embeddings.

To ground reasoning in real-world perspectives, we retrieve individuals demographically similar to d_q using dense embedding retrieval. We encode demographic descriptions with an embedding model and rank respondents in I by embedding similarity to d_q . The top- K individuals form the demographic set R_q ,

$$R_q = \{i_1, i_2, \dots, i_K\}, \quad (9)$$

with corresponding value profiles

$$V_q = \{V_i \mid i \in R_q\}. \quad (10)$$

3.2.2. MULTI-VALUE PERSONA AGENT SIMULATION

Given V_q , we instantiate Value-Persona Agents that simulate culturally grounded reasoning under ontology constraints. We denote the agent as G_{persona} and instantiate it per retrieved individual $i \in R_q$. For convenience, index the retrieved ontology triples as $O_q = \{t_h\}_{h=1}^{|O_q|}$ and let $C_q = \{c_a, c_b \mid t_h \in O_q\}$ denote the set of sub-classes referenced by the retrieved ontology triples. Concretely, each agent is conditioned on the ontology context O_q (typically 3–9 triples), the individual’s value summaries restricted to C_q , and demographic attributes d_i . We denote the filtered profile as $V_{i,q} = \{s_i(c) \mid c \in C_q\}$. The agent-specific conditioning context is:

$$z_i = \text{Concat}(O_q, V_{i,q}, d_i). \quad (11)$$

Here z_i denotes the conditioning context for the persona associated with individual i .

Simulated reasoning trace. Given query q , each agent generates (i) an answer and (ii) an explicit simulated reasoning trace that forms a coherent chain of thought under the provided evidence and ontology constraint:

$$G_{\text{persona}}(q, z_i) = (\hat{y}_i, \rho_i). \quad (12)$$

Here $\hat{y}_i$ denotes the agent’s answer and ρ_i denotes its natural-language reasoning trace. We collect outputs across all retrieved individuals and ontology triples and denote the set by A :

$$A = \{G_{\text{persona}}(q, z_i) \mid i \in R_q\}. \quad (13)$$

3.2.3. ONTOLOGY-GUIDED FINAL JUDGMENT VIA CONSTRAINED META-ADJUDICATION

A Final Judgment Agent G_{judge} synthesizes the final prediction by performing *Constrained Meta-Adjudication* over the candidate outputs from Value-Persona Agents. Given the multi-agent set A and the query q (question and response options), the judge outputs:

$$\hat{y}_q = G_{\text{judge}}(A, q). \quad (14)$$

Here $\hat{y}_q$ denotes the final prediction for query q . The judge does not receive O_q or $\mathcal{V}_q$ directly; ontology and profile grounding are carried through the persona outputs.

Compared to majority voting, G_{judge} uses a constrained, evidence-first protocol. The Judgment Agent follows an evidence-first adjudication protocol that prioritizes grounded support and ontology consistency before consulting vote signals:⁶

1. **Evidence & consistency.** For each $(\hat{y}_i, \rho_i)$, score grounding of ρ_i and ontology compliance, then aggregate scores per option y .
2. **Vote as secondary cue.** If the leading options have comparable evidence strength, consult a Vote summary as a secondary signal (otherwise ignore it).
3. **Relevance tie-break.** If still tied, choose the option supported by personas more relevant to d_q .

This adjudication is conducted within a single LLM call, with the above criteria guiding the judge’s internal reasoning rather than operating as separate rule-based modules.

3.3. Implementation Details

Our implementation consists of: (i) a fixed ontology **taxonomy** with 12 top-classes and 64 sub-classes; (ii) a **Summarization Agent** G_{sum} , an LLM-based summarizer that generates category-specific summaries $s_i(c)$ from raw respondent responses $\mathcal{R}_i$ under a “no-new-concepts” constraint; (iii) a **Topic-Selection Agent** G_{topic} , a pretrained text encoder fine-tuned on WVS with top-class supervision, which selects the top- k top-classes; (iv) a **category selection** policy that retains the top-3 fine-grained value categories per query; (v) an **ontology triple retrieval** step that performs dense retrieval over ontology triples for each selected category and returns up to the top-3 triples per category; and (vi) a **persona retrieval setting** where the default number of retrieved individuals (personas) is set to $K=5$.

⁶Mechanism-level cases distinguishing OG-MAR from ensemble-style aggregation are provided in Appendix G.5.

4. Experimental Design and Setup

4.1. Setup

Models Used. We use GPT-4o-mini (Achiam et al., 2023) and Gemini 2.5 (Google DeepMind, 2025) via APIs, and Qwen 2.5 (Team, 2024) and EXAONE 3.5 (An et al., 2024) as open-source models for the generation task in both our Persona Agent and Final Judgment Agent. We also used GPT-o4-mini (OpenAI, 2025) for object properties construction and values profile generation. To ensure stable behavior, we set the temperature to 0 across all models.

We use dense embedding retrieval with E5-base embeddings (Wang et al., 2022a). For demographic retrieval, we encode the target demographic description and each respondent’s demographic profile, then rank respondents by embedding similarity to obtain the top- K demographically similar individuals. For ontology retrieval, we embed ontology triples and retrieve the top- M triples by similarity to the query.

Additionally, for topic classification, we fine-tune DeBERTa-v2-xxlarge (He et al., 2020) on WVS data for 3 epochs (batch size=4, learning rate= 5×10^{-6}), following optimized configurations for value identification (Kiesel et al., 2023; Balikas, 2023)⁷.

4.2. Datasets

Retrieval Corpus We use the *World Values Survey* (WVS) (Haerpfer et al., 2022) as the retrieval corpus. WVS is a large-scale cross-national survey of human values and socio-cultural attitudes with structured demographic attributes (e.g., country, age, gender, education). In our setting, we use predefined 12 topics in the WVS⁸, which provide a globally diverse and publicly available source of value-related responses and enable consistent retrieval of relevant demographic evidence for downstream inference.

Test Datasets To evaluate generalization beyond the retrieval corpus, we use six regional social-survey datasets with value-related questions and WVS-comparable demographic metadata: **EVS** (Europe), **GSS** (U.S.), **CGSS** (China), **ISD** (India), **LAPOP** (Latin America and the Caribbean), and **Afrobarometer** (Africa). We use clustering-based sampling to select 2,000 representative instances as test data, for efficient yet balanced evaluation across diverse datasets⁹.

⁷Detailed training analyses are provided in Appendix A.

⁸The 12 topics are provided in Appendix C, Table 9.

⁹Detailed dataset descriptions, sampling process, and an additional post-cutoff experiment are provided in Appendix C.

4.3. Evaluation Metrics

We evaluate model performance using both item-level and distribution-level metrics. For item-level evaluation, we report binary accuracy, where each prediction is scored as correct or incorrect after applying the corresponding response-type rule. This yields a unified 0/1 correctness criterion across datasets while retaining item-specific answer formats. We use four response types:

- **Two-choice binary items:** two-option questions, evaluated by exact matching between the predicted label and the gold label.
- **Derived binary items:** multiple-choice questions mapped into two fixed semantic buckets; a prediction is correct if the predicted and gold labels fall into the same bucket.
- **Ordinal items:** ordered response items. For accuracy, raw responses are binarized using the midpoint of the original scale, and compared with the gold label.
- **Nominal exact-match items:** unordered multi-choice questions, evaluated by exact matching between the predicted label and the gold label.

For ordinal items, we additionally report Mean Absolute Error (MAE) on the valid ordinal scale:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (15)$$

Here, y_i and $\hat{y}_i$ denote the gold and predicted ordinal values after applying the item-specific valid response range.¹⁰

For distribution-level evaluation, we compare the predicted and gold survey response distributions using EMD, Proportion Correlation, and TVD.¹¹

4.4. Baselines

We compare against two single-pass prompting baselines, one single-agent aggregation baseline, and two multi-step baselines spanning multi-agent deliberation and retrieval augmentation: (1) **Zero-shot**, a single prompt without scaffolding; (2) **Role Assignment** (Tao et al., 2024), which conditions generation on an explicit culturally grounded role; (3) **Self-consistency** (Wang et al., 2022b), which samples multiple reasoned outputs from the same model and takes a majority vote over them; (4) **Debate** (Ki et al., 2025), a multi-agent framework with iterative critique and refinement; and (5) **ValuesRAG** (Seo et al., 2025), which grounds

generation in retrieved survey evidence. For a fair comparison, we use the same retrieval setting for *ValuesRAG* in the main experiments¹².

5. Experimental Results

Table 1 reports results on six regional benchmarks and four Large Language Model backbones, showing that baselines offer only incremental gains and fail under cultural distribution shift. **Zero-shot** Inference is efficient but often reverts to culture-default priors, yielding inconsistent value-sensitive judgments across regions. **Role Assignment** adds culturally framed prompting, yet remains weakly grounded in empirically observed value distributions and is sensitive to prompt formulation. **Self-Consistency** improves robustness via sample aggregation, but it does not enforce demographic grounding or conceptual coherence and may amplify majority bias. **Debate** introduces critique-and-refinement, but without explicit evidence constraints it is prone to drift and still fails to capture cross-topic value dependencies. **ValuesRAG** is competitive by grounding generation in survey evidence but treats values as unstructured snippets, which restricts control over structured relationships.

In contrast, **OG-MAR** integrates ontology-guided triple retrieval and multi-persona simulation with a judgment agent that explicitly weighs ontology consistency and demographic proximity. This design yields strong and consistent performance across regions, achieving average accuracies of 0.6308 on Gemini 2.5 Flash Lite, 0.5705 on Qwen 2.5, and 0.6317 on EXAONE, while remaining competitive on GPT-4o-mini with an average accuracy of 0.6007. Notably, **OG-MAR** delivers particularly large gains on culturally challenging settings such as CGSS and ISD, suggesting that structured cultural relations and demographically grounded personas are most beneficial when the target distribution deviates from dominant pretraining priors.

5.1. Ablation Studies

5.1.1. VARYING THE NUMBER OF RETRIEVED INDIVIDUALS

We investigate the impact of retrieval size K on **OG-MAR**’s performance by varying the number of retrieved demographically similar individuals across $K \in \{1, 3, 5, 10\}$. Figure 3 reports average accuracy for GPT-4o-mini, Gemini 2.5 Flash Lite, QWEN 2.5, and EXAONE 3.5.

All four models achieve their best overall performance at $K=5$, outperforming other retrieval sizes by +0.003 to +0.07 across different models. While Gemini 2.5 Flash Lite shows minimal difference between $K=3$ and $K=5$, the re-

¹⁰Average MAE scores across regional datasets are shown in Figure 6.

¹¹Detailed evaluation results are provided in Appendix B.

¹²Detailed baseline explanations and an additional post-training comparison are provided in Appendix D.

Table 1. Accuracy of baseline methods across regional datasets. Bold text indicates the best performance, underlined text the second-best performance. * denotes significant improvements (paired t -test with Holm-Bonferroni correction, $p < 0.05$) over all baselines. † denotes our proposed method.

Method	EVS (Europe)	GSS (United States)	CGSS (China)	ISD (India)	AFRO (Africa)	LAPOP (Latin America)	Avg.
GPT-4o mini							
Zero-shot	0.5606	0.5164	0.5847	0.6139	0.5324	0.5760	0.5640
Role (2024)	0.5892	0.5184	<u>0.6014</u>	0.6060	<u>0.5505</u>	0.5674	0.5722
Self-consistency (2022b)	0.5558	0.4920	0.5631	0.5976	0.5224	0.5551	0.5477
Debate (2025)	0.5985	<u>0.5509</u>	0.5993	0.6568	0.5343	0.5306	0.5784
ValuesRAG (2025)	0.6127	0.5589	0.5889	0.6420	0.5654	0.6085	0.5961
OG-MAR (Ours)†	0.6206*	0.5480	0.6509*	0.6192	0.5389	0.6268	0.6007*
Gemini 2.5 Flash Lite							
Zero-shot	0.5681	0.4957	0.6467	0.5000	0.5282	0.6225	0.5602
Role (2024)	0.5786	0.4992	<u>0.6669</u>	0.5521	0.5313	0.5852	0.5689
Self-consistency (2022b)	0.5489	0.4728	0.6063	0.4705	0.5182	<u>0.6268</u>	0.5406
Debate (2025)	0.5977	0.5138	0.6348	<u>0.6335</u>	0.5046	0.5331	0.5696
ValuesRAG (2025)	<u>0.6075</u>	<u>0.5376</u>	0.6084	0.6041	<u>0.5472</u>	0.5339	<u>0.5731</u>
OG-MAR (Ours)†	0.6249*	0.5489*	0.7017*	0.7007*	0.5701*	0.6385*	0.6308*
QWEN 2.5							
Zero-shot	0.5199	0.5069	0.2704	<u>0.7222</u>	0.4814	0.4908	0.4986
Role (2024)	0.5357	0.5037	0.3463	0.7452	<u>0.5014</u>	0.4712	0.5172
Self-consistency (2022b)	0.5096	0.4975	0.3289	0.6278	0.4080	0.4975	0.4782
Debate (2025)	0.5511	0.5174	0.4578	0.6320	0.4875	0.4332	0.5132
ValuesRAG (2025)	<u>0.5538</u>	<u>0.5215</u>	<u>0.4697</u>	0.6591	0.4724	<u>0.5268</u>	<u>0.5339</u>
OG-MAR (Ours)†	0.5898*	0.5325*	0.5220*	0.6599	0.5180	0.6005	0.5705*
EXAONE 3.5							
Zero-shot	0.5143	0.5311	0.2885	0.6041	0.4054	0.5006	0.4740
Role (2024)	0.5319	0.5326	0.3129	0.6048	0.4077	0.4602	0.4750
Self-consistency (2022b)	0.5490	0.5266	0.2697	0.6122	0.4086	0.5368	0.4838
Debate (2025)	0.5713	0.5407	0.5624	<u>0.6773</u>	0.4995	0.4939	0.5575
ValuesRAG (2025)	0.5172	<u>0.5520</u>	<u>0.5833</u>	0.6446	0.4794	<u>0.5913</u>	<u>0.5613</u>
OG-MAR (Ours)†	0.6080*	0.5636	0.6307*	0.7810*	0.5045*	0.7022*	0.6317*

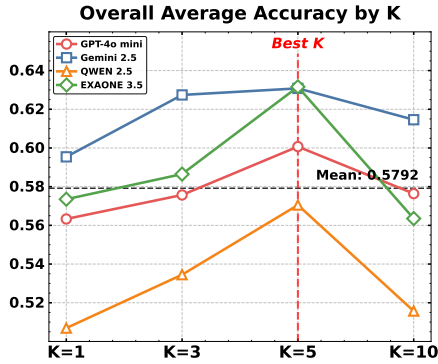


Figure 3. Performance comparison of four models across $K \in \{1, 3, 5, 10\}$ on average. Red dashed vertical lines indicate the best K and gray horizontal lines show the overall mean accuracy.

maining models exhibit substantially larger improvements from $K=3$ to $K=5$, with gains ranging from 0.03 to 0.05. When retrieval size increases to $K=10$, all models show clear performance degradation, with accuracy drops ranging from 0.02 to 0.07 compared to $K=5$. These results reveal a clear trade-off: $K=1$ retrieves narrow value-persona signals, whereas $K=5$ offers richer yet stable context. Consequently, we adopt $K=5$ as the default retrieval depth throughout our experiments¹³.

¹³Complete results for all regional datasets are in Appendix I.1.

5.1.2. IMPACT OF VALUE INFERENCE GENERATION

To assess the impact of value inference generation, we compare our two-step process (with K personas and a judgment agent) against a similar two-step process where a single Value Inference Agent replaces the K persona agents by first inferring a value profile for the target individual, after which the same judgment agent answers questions based only on this profile. We evaluate both approaches across all six regional datasets using the same retrieval and setup as in our main experiments. Figure 4 presents the accuracy comparison between the two architectures.

Among the four models, GPT-4o-mini slightly benefits from the Value Inference Variant by around 0.01 on average, particularly showing higher accuracy on GSS (+0.07) and CGSS (+0.11). In contrast, the other three models achieve consistent gains with **OG-MAR**, ranging from +0.03 to +0.10 on average across datasets. While the Value Inference Variant outperforms **OG-MAR** on GSS across all four models, **OG-MAR** achieves clearly higher accuracy on the remaining datasets for most models. These results suggest that explicitly simulating multiple personas and preserving their distinct value profiles provides the judgment model with richer and more diverse evidence¹⁴, which in turn al-

¹⁴Appendix G.4 provides a qualitative case of preserving competing cultural cues through multi-persona simulation.

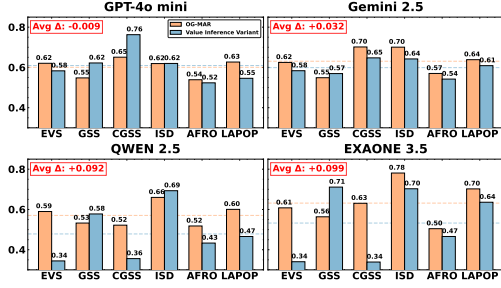


Figure 4. Performance comparison between OG-MAR and the Value Inference Variant. Accuracy over four models on six regional datasets. Dashed lines show per-method average accuracy; red boxes report the average gap (Avg Δ = OG-MAR - Variant).

Table 2. Average accuracy of the full OG-MAR framework and the single-judge variant across four LLMs.

Model	Method	Avg. Accuracy
GPT-4o mini	OG-MAR	0.6007
	Single-Judge	0.5987
Gemini 2.5	OG-MAR	0.6308
	Single-Judge	0.6022
QWEN 2.5	OG-MAR	0.5705
	Single-Judge	0.5311
EXAONE 3.5	OG-MAR	0.6317
	Single-Judge	0.5627

lows **OG-MAR** to maintain stronger performance than the Value Inference Variant in most models and datasets.

5.1.3. IMPACT OF MULTI-PERSONA REASONING

We compare the full **OG-MAR** framework with a single-judge variant that skips persona simulation and directly generates the final answer from the same inputs (K individuals’ demographics, value summaries, and ontology context). Using the setup of Section 5.1.2, we evaluate both variants on all datasets; Table 2 reports the resulting accuracies.

Overall, **OG-MAR** achieves higher average accuracy across all four models (+0.002 on GPT-4o mini, +0.03 on Gemini 2.5 Flash Lite, +0.04 on QWEN 2.5, and +0.07 on EXAONE 3.5). Although the single-judge variant is consistently better on GSS, **OG-MAR** improves most other datasets¹⁵. These results suggest that persona simulation is a meaningful contributor to performance, particularly when regional benchmarks require reconciling multiple, potentially competing value considerations. This effect is especially pronounced when the input demographic set induces diverse value profiles. At the same time, the single-judge model remains competitive, implying that **OG-MAR**’s gains are not driven solely by the simulation layer. Instead, improvements also stem from the shared ontology-grounded retrieval and value

¹⁵Detailed per-dataset results are provided in Appendix I.2.

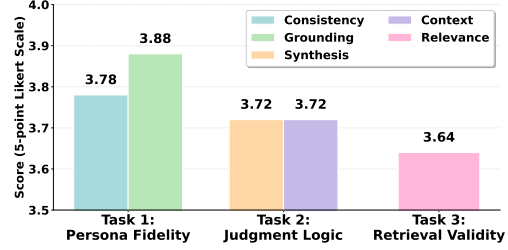


Figure 5. Average human evaluation scores (5-point Likert scale) across three tasks: Persona Fidelity (Consistency, Grounding), Judgment Logic (Synthesis, Context), and Retrieval Validity (Relevance). Scores are averaged over nine expert raters.

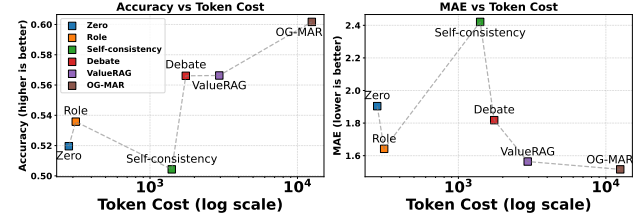


Figure 6. Performance-cost trade-off across methods. Left: accuracy vs total tokens (higher is better). Right: MAE vs total tokens (lower is better). Markers denote methods; the dashed line shows performance changes as token usage increases.

summarization pipeline, which provides structured, survey-backed evidence for downstream reasoning¹⁶.

6. Discussion

Qualitative Analysis To complement quantitative results, we conducted a human evaluation of **OG-MAR**’s reasoning traces with nine domain experts using three 5-point Likert tasks: *Persona Fidelity* (Task 1), *Judgment Logic* (Task 2), and *Retrieval Validity* (Task 3).

Overall, **OG-MAR** shows consistent interpretability across six regional datasets. Notably, it achieves the highest *Grounding* on **CGSS (China)** (4.02), slightly exceeding **GSS (U.S.)** (3.97), suggesting ontology-guided value injection mitigates “culture-default” tendencies by encouraging evidence-based reasoning. The Judgment Agent also attains strong *Synthesis Logic* (Avg. 3.72), indicating evidence-first adjudication remains effective across culturally distinct regions, while ontology retrieval maintains high relevance (Avg. 3.64), supporting that retrieved triples provide meaningful evidence for downstream reasoning.

Qualitatively, experts frequently noted that persona rationales remained consistent with the retrieved value summaries and that final decisions explicitly reconciled conflicting value signals rather than defaulting to generic responses. The case studies further illustrate how evidence-first adju-

¹⁶A component-level ablation study is provided in Appendix I.4.

dication can correct or down-weight noisy persona claims when they are not supported by ontology-linked evidence. Remaining failures are typically attributable to sparse ontology coverage or imperfect demographic retrieval, which can limit grounding even when the judgment logic is coherent.¹⁷

Performance–Token Usage Trade-off. Figure 6 shows the trade-off between token cost and prediction quality across methods. We measure cost by the total number of input and output tokens, and use a log-scale x-axis to highlight differences in usage. Performance is reported with Accuracy (higher is better) and MAE (lower is better). Single-pass prompting baselines lie in the lowest-cost regime but generally achieve weaker performance. RAG method baseline (ValuesRAG) and multi-step methods that retrieve external context or aggregate multiple samples consume substantially more tokens and typically improve quality.

OG-MAR incurs the highest token budget, however, this cost reflects its intended role: not as a token-efficient replacement for lightweight prompting, but as a structured reasoning framework that organizes cultural values through ontology-guided relations, explicitly separates cultural dimensions, and supports transparent multi-agent adjudication. Its higher accuracy and lower average MAE therefore suggest that the additional computation contributes to more grounded and interpretable cultural reasoning, rather than merely adding inference overhead.

7. Conclusion

In this paper, we presented **OG-MAR**, an ontology-guided multi-agent framework for culturally aligned LLM inference. **OG-MAR** integrates CQ-driven cultural ontology construction with demographically grounded retrieval from WVS, and combines value-persona simulation with a judgment agent that enforces ontology consistency. Across six regional benchmarks and four LLM backbones, **OG-MAR** improves cultural alignment and robustness over competitive baselines. Beyond quantitative gains, our qualitative and ablation analyses help explain when and why **OG-MAR** works, highlighting the complementary roles of ontology-grounded retrieval, value summarization, and multi-persona reasoning in producing more interpretable outputs.

8. Limitations

While our results show that **OG-MAR** improves cultural alignment across diverse regions and LLM backbones, it does not guarantee optimal behavior for every query, demographic subgroup, or survey domain. Also, the performance

can be sensitive to the underlying model, prompt context, and retrieval noise. In particular, cross-domain generalization remains limited: regional surveys are not always fully aligned with the WVS-derived taxonomy, and topic selection may fail when questions use domain-specific phrasing or reflect culturally local constructs.

Moreover, demographic retrieval may miss minority or underrepresented profiles, and imperfect ontology triples can propagate structured but inaccurate constraints into downstream reasoning. The multi-agent simulation also increases inference cost and latency relative to single-pass baselines.

Future work will address these limitations by: (1) developing more robust verification mechanisms for ontology triple accuracy and consistency; (2) improving cross-domain topic classification and uncertainty-aware retrieval to better cover underrepresented groups and survey-specific phrasing; and (3) designing efficient routing or pruning techniques within the multi-agent simulation to reduce computational cost without sacrificing alignment quality.

Impact Statement

While **OG-MAR** aims to reduce cultural misalignment, its demographic retrieval and survey-grounded cultural signals raise ethical risks related to profiling, bias, and fairness. In high-stakes settings, demographic-conditioned outputs may reinforce stereotypes, encode survey artifacts, or amplify historical biases. **OG-MAR** is intended to surface value considerations for scrutiny and auditing, not to justify demographic generalizations.

Researchers and practitioners should evaluate performance across subpopulations (including minority and underrepresented groups), monitor for harmful pattern reproduction, and apply strong privacy and data-governance safeguards (including protections against re-identification). We recommend using **OG-MAR** only within accountable oversight workflows; future work should study uncertainty communication, debiasing, and safe deployment protocols.

Code and Model Availability

Code and sample data are available in the [project repository](#). The repository also includes scripts, prompts, and instructions to reproduce the main experiments. The topic classifier is available on [Hugging Face](#).

Acknowledgements

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¹⁷Detailed qualitative evaluation tasks, procedures, and results are provided in Appendix E; qualitative case studies, including evidence-first adjudication cases, are provided in Appendix G.

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A. Training Details and Loss Curves

A.1. DeBERTa-v2-xxlarge Fine-tuning

We fine-tuned DeBERTa-v2-xxlarge (He et al., 2020) on the WVS category classification task (191 questions, 12 categories) using batch size 4, learning rate 5×10^{-6} , AdamW optimizer (weight decay 0.01), and FP16 precision on an NVIDIA A100 GPU. Training terminated early at epoch 3 when validation Top-3 accuracy reached 100%, as shown in Figure 7.

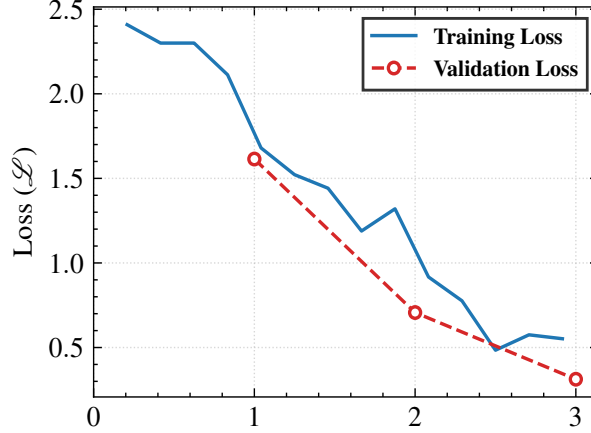


Figure 7. **Training and validation loss curves for DeBERTa-v2-xxlarge fine-tuning.** The x-axis represents epochs. Training loss (blue solid line) exhibits minor fluctuations typical of small-batch optimization, while validation loss (red dashed line, evaluated every 48 steps) decreases monotonically from 1.61 to 0.31 across three epochs, indicating effective learning without overfitting.

The early stopping criterion based on Top-3 accuracy ensures computational efficiency. Despite minor fluctuations in training loss during the final epoch, validation loss decreases consistently across all epochs on the WVS dataset.

A.2. Value Category Classification

Table 3. **Topic classification performance on six regional test datasets and WVS validation data.** Top- k : fraction of questions where the true category appears in top- k predictions. All metrics in $[0,1]$.

Dataset	Top-1	Top-2	Top-3
Afrobarometer	0.5037	0.6875	0.7574
CGSS	0.3375	0.5079	0.6656
EVS	0.4315	0.5560	0.6680
GSS	0.4545	0.6667	0.7765
ISD	0.5439	0.7071	0.7950
LAPOP	0.4396	0.6577	0.7349
WVS (val)	0.9583	1.0000	1.0000

The lower Top-1 accuracy on the six regional test datasets reflects taxonomy coverage limitations and question-style mismatch rather than classifier failure. Regional surveys were designed independently of WVS, and their phrasings do not always align with the 12-category taxonomy on which the model was trained. Ground-truth labels were assigned through independent blinded consensus annotation to ensure reliability. To account for this uncertainty, **OG-MAR** retrieves ontology triples from all three predicted categories rather than relying on the top prediction alone, broadening coverage when the correct category may not rank first. Top-3 accuracy therefore directly corresponds to the operational condition under which the framework runs. We examine how topic-classifier accuracy affects downstream performance in Appendix A.3.

A.3. Topic-Classifier Error Propagation

In **OG-MAR**, DeBERTa’s predicted topic categories determine which ontology triples are retrieved for each question. To measure whether classifier errors propagate to downstream performance, we partitioned the regional test questions into two

groups based on whether the ground-truth category appeared in DeBERTa’s Top-3 predictions.

Table 4. OG-MAR average scores by topic-classifier coverage. “In Top-3” indicates that the ground-truth category appears in DeBERTa’s Top-3 predictions. “Not in Top-3” indicates it does not. Evaluation metric is Avg. Score. Bold denotes the higher value per model.

Model	In Top-3	Not in Top-3
GPT-4o mini	0.6157	0.6114
Gemini 2.5 Flash Lite	0.6435	0.6232
QWEN 2.5	0.7039	0.6448
EXAONE 3.5	0.6317	0.6149

Table 4 reports average scores for each group across four backbone models. The **In Top-3** group consistently achieves higher scores than the **Not in Top-3** group, though the gap varies across backbones. GPT-4o mini shows the smallest difference at 0.004, while QWEN 2.5 shows the largest at 0.059. Gemini 2.5 Flash Lite and EXAONE 3.5 fall in between at 0.020 and 0.017, respectively. Despite this backbone-dependent variation, the pipeline continues to produce meaningful outputs even when the correct category is not covered. We attribute this robustness to the semantic proximity of neighboring categories and the partial relevance of triples retrieved from adjacent topics.

B. Additional Evaluation Metrics

To complement the main evaluation, we conduct additional distribution-level analyses using three metrics: Earth Mover’s Distance (EMD; lower is better), Proportion Correlation (higher is better), and Total Variation Distance (TVD; lower is better). These metrics evaluate whether the simulated survey responses match the original ordinal response distributions, rather than relying only on a coarse binary outcome. As summarized in Table 5, the results are averaged over six survey datasets: EVS, GSS, CGSS, ISD, AFRO, and LAPOP.

Table 5. Additional distribution-level evaluation across four LLM backbones. All scores are averaged over six survey datasets. **Bold** indicates the best score within each backbone and metric. **OG-MAR** achieves the best EMD, Proportion Correlation, and TVD on GPT-4o mini, Gemini 2.5 Flash Lite, and QWEN 2.5, showing consistent distributional alignment beyond the main evaluation metric.

Method	EMD ↓	Prop. Corr. ↑	TVD ↓
GPT-4o mini			
Zero-shot	0.6744	0.4787	0.4175
Role	0.6559	0.4082	0.4230
Self-consistent	0.7760	0.4136	0.4511
Debate	0.6609	0.4494	0.4179
ValuesRAG	0.6800	0.4651	0.4370
OG-MAR	0.6203	0.4941	0.3950
Gemini 2.5 Flash Lite			
Zero-shot	0.7169	0.4340	0.4410
Role	0.7416	0.4137	0.4471
Self-consistent	0.7997	0.4547	0.4831
Debate	0.6087	0.4853	0.4173
ValuesRAG	0.6789	0.4834	0.4278
OG-MAR	0.5888	0.5205	0.4023
QWEN 2.5			
Zero-shot	0.8500	0.3474	0.5277
Role	0.7670	0.3698	0.4972
Self-consistent	0.8003	0.3880	0.5045
Debate	0.7355	0.3508	0.4630
ValuesRAG	0.7531	0.4029	0.4960
OG-MAR	0.6992	0.4264	0.4606
EXAONE 3.5			
Zero-shot	0.9159	0.2658	0.5572
Role	0.8678	0.2575	0.5324
Self-consistent	0.9784	0.2776	0.5556
Debate	0.7940	0.2943	0.5342
ValuesRAG	0.8540	0.2969	0.5030
OG-MAR	0.8014	0.3052	0.4894

Table 5 shows that **OG-MAR** achieves the best EMD, Proportion Correlation, and TVD on GPT-4o mini, Gemini 2.5 Flash Lite, and QWEN 2.5. This indicates that **OG-MAR** improves distributional alignment across multiple backbones, not merely under a single evaluation metric. The only partial exception appears on EXAONE 3.5, where **OG-MAR** still obtains the

Table 6. **Distributional evaluation of OG-MAR under binary and vote-proportion variants.** The vote-proportion variant uses pre-aggregation persona votes and is not directly comparable to binary baselines. Nevertheless, it generally improves distributional recovery, especially in TVD.

OG-MAR Variant	GPT	Gemini	QWEN	EXAONE
TVD ↓				
Binary	0.3950	0.4023	0.4606	0.4894
Vote prop.	0.2057	0.2271	0.2473	0.2985
Prop. Corr. ↑				
Binary	0.4941	0.5205	0.4264	0.3052
Vote prop.	0.5645	0.4925	0.4871	0.4569

best Proportion Correlation and TVD, but does not achieve the best EMD. This suggests that the effectiveness of **OG-MAR** remains generally robust, while weaker instruction-following backbones may limit some aspects of distributional recovery.

We further examine whether **OG-MAR**’s pre-aggregation persona vote proportions provide a more fine-grained approximation of the target response distribution. Table 6 compares the standard binary prediction setting with the vote-proportion setting. The binary setting uses the final hard prediction from the Judgment Agent and is directly comparable to all baselines. In contrast, the vote-proportion setting uses the pre-aggregation votes from $K = 5$ persona agents as a continuous approximation of $P_{\text{model}}(y = 1 \mid q, g)$; therefore, it is reported separately and should not be interpreted as a direct baseline comparison.

As shown in Table 6, the vote-proportion variant substantially improves TVD across all four backbones, suggesting that the intermediate persona votes contain useful distributional information before final binary aggregation. This result further supports the design motivation of **OG-MAR**: multi-agent persona reasoning can capture diverse response tendencies that are partially lost when converted into a single hard prediction.

C. Dataset Details

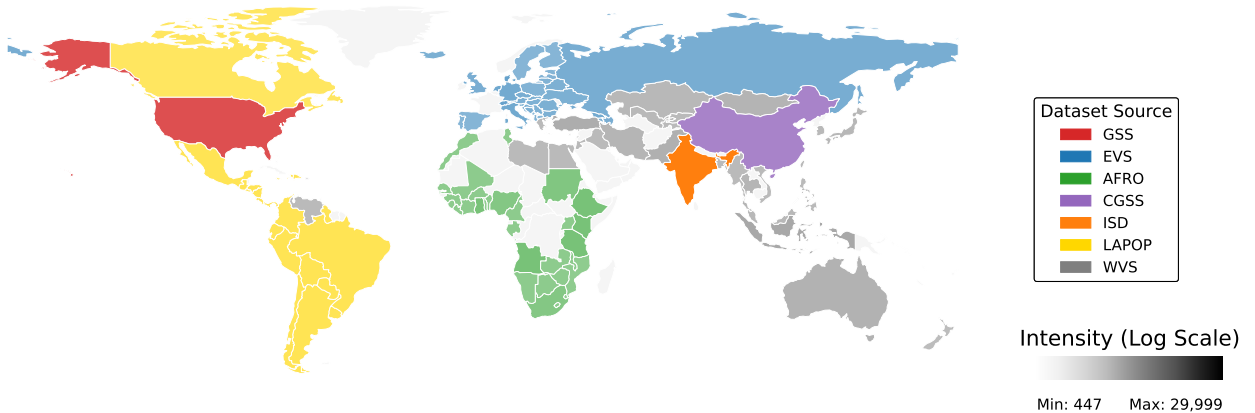


Figure 8. **Geographic coverage of cultural value datasets used in this study.** Each country is colored according to its primary data source, prioritizing regional surveys over the global World Values Survey. Regional datasets include the General Social Survey for the United States, the European Values Study for Europe, Afrobarometer for Africa, the Chinese General Social Survey for China, and the India Social Dataset for India. Countries without regional coverage are represented by WVS data shown in gray. Color intensity reflects participant count on a logarithmic scale, ranging from 447 to 29,999 respondents per country. This multimodal approach ensures both regional specificity and global breadth in cultural alignment research.

Table 7. **Summary statistics of the retrieval corpus (WVS) and six test datasets.** For test datasets, “#Value Qs” denotes the value-related items retained after our preprocessing/topic mapping (not necessarily the full questionnaire length).

Dataset	Type	Region	Wave / Year	#Countries	#Respondents	#Value Qs
<i>Retrieval Corpus</i>						
WVS (World Values Survey)	Retrieval	Global	2017–2022	64	94,728	239
<i>Test Datasets</i>						
EVS (European Values Study)	Test	Europe	2017	–	59,438	211
GSS (General Social Survey)	Test	United States	2021–2022	–	8,181	44
CGSS (Chinese General Social Survey)	Test	China (E. Asia)	2021	–	~8,148	58
ISD (Pew India Survey Dataset)	Test	India (S. Asia)	2019–2020	–	29,999	33
LAPOP (AmericasBarometer)	Test	Latin America	2021	–	64,352	48
Afrobarometer	Test	Africa	2022	–	~48,100	144

Table 8. **Data sources used in our experiments.** We use the World Values Survey (WVS) as the retrieval corpus, and evaluate generalization on six external test datasets (EVS, GSS, CGSS, ISD, LAPOP, and Afrobarometer), with official access links provided for reproducibility.

Dataset	Link
<i>Retrieval Corpus</i>	
WVS	https://www.worldvaluessurvey.org/wvs.jsp
<i>Test Datasets</i>	
EVS (European Values Study)	https://europeanvaluesstudy.eu
GSS (General Social Survey)	https://gss.norc.org
CGSS (Chinese General Social Survey)	https://cgss.ruc.edu.cn
ISD (Pew India Survey Dataset)	https://www.pewresearch.org/dataset/india-survey-dataset/
LAPOP (AmericasBarometer)	https://www.vanderbilt.edu/lapop
Afrobarometer	https://www.afrobarometer.org

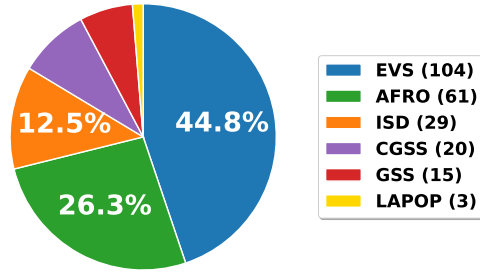


Figure 9. **Distribution of selected value questions across regional datasets.**

To ensure the quality and relevance of our value questions, we employed a systematic selection process for the regional datasets. First, we considered the actual response rates from our extracted representative samples (see Section C.2) and retained the top 80% of questions with valid responses in each dataset. This ensured sufficient data coverage for meaningful analysis. Second, we excluded questions that were overly dependent on personal circumstances and behaviors, such as “Did your household ever run out of water?”, “In the past year, how often have you used radio?”, “Are you a member of a trade union?”, or “Based on your experience, how easy or difficult is it to find out how government uses tax revenues?”. These questions tend to reflect personal circumstances rather than underlying values. Third, we excluded questions requiring knowledge of specific countries or domestic institutions, such as “To what extent do you think [country] is democratic?” or “How much do you trust the Electoral Commission of Ghana?”. It would be difficult to generalize these questions across cultural contexts. After applying these criteria, we obtained a final set of questions distributed across the six regional datasets. Figure 9 shows the distribution of these selected questions across datasets.

Table 9. **Distribution of Values-related Questions in WVS.** The questions were categorized into 12 topics with a total of 250 questions in the original WVS questionnaire covering most of the dimensions of values.

Topic	Count
Social Values, Norms, Stereotypes	45
Happiness and Wellbeing	11
Social Capital, Trust and Organizational Membership	47
Economic Values	6
Perceptions of Corruption	9
Perceptions of Migration	10
Perceptions of Security	21
Perceptions about Science and Technology	6
Religious Values	12
Ethical Values	23
Political Interest and Political Participation	35
Political Culture and Political Regimes	25

C.1. Dataset Information

Retrieval Corpus: World Values Survey (WVS). We use the *World Values Survey* (WVS) as our retrieval corpus. In this setting, WVS covers 64 countries/territories with 94,728 respondents and a 290-item common questionnaire. We organize the value space into 12 predefined topics (Table 9) and, after preprocessing, retain 239 region-agnostic ordinal value items for retrieval-augmented inference.

Test Datasets. To evaluate performance of **OG-MAR**, we use six regional social-survey datasets spanning East Asia, South Asia, Europe, North America, Latin America, and Africa. For each test dataset, the reported number of value questions refers to the subset retained after our preprocessing/topic mapping (not necessarily the full questionnaire length).

EVS (European Values Study). EVS is a cross-national survey program designed to measure human values and socio-cultural attitudes across Europe using harmonized instruments. We use the EVS 2017 integrated dataset, which combines nationally representative samples from 36 countries (59,438 respondents) collected during the 2017–2021 fieldwork period. In our evaluation, we retain 211 value-related items under our preprocessing and topic mapping, enabling within-Europe generalization tests under relatively consistent survey design and documentation.

GSS (General Social Survey). GSS is a long-running repeated cross-sectional survey for the United States, featuring a replicating core plus rotating topical modules and rich demographics. We use the 2021 and 2022 cross-sections (4,032 and 4,149 completes; 8,181 total), which reflect a major methodological transition (e.g., 2021 push-to-web design and 2022 mixed-mode transition). We retain 44 value-related items, making GSS a useful stress test for robustness under both cultural shift (vs. the global WVS corpus) and survey-mode/design differences across waves.

CGSS (Chinese General Social Survey). CGSS is a nationally representative household survey measuring social attitudes and values in China, accompanied by detailed demographic covariates. We use the 2021 release with 8,148 valid samples (often documented as being drawn nationwide across many communities/provinces). After preprocessing and topic alignment, we retain 58 value-related items, providing a linguistically and institutionally distinct setting for cross-cultural generalization beyond the WVS retrieval corpus.

ISD (Pew India Survey Dataset). To represent South Asia, we use Pew Research Center’s India Survey Dataset, a nationally representative face-to-face survey administered from Nov. 2019 to Mar. 2020. The dataset contains 29,999 adult interviews and covers broad attitudinal themes (e.g., identity, nationalism, tolerance), alongside demographics and survey weights. We retain 33 value-related items under our mapping, supporting evaluation in a highly heterogeneous population with strong methodological transparency.

LAPOP (AmericasBarometer). LAPOP is a large-scale cross-national public opinion survey program in Latin America that measures citizens’ political attitudes, democratic governance, institutional trust, and socio-economic perceptions with harmonized instruments and rich demographics. We use the 2021 AmericasBarometer release (64,352 respondents) and,

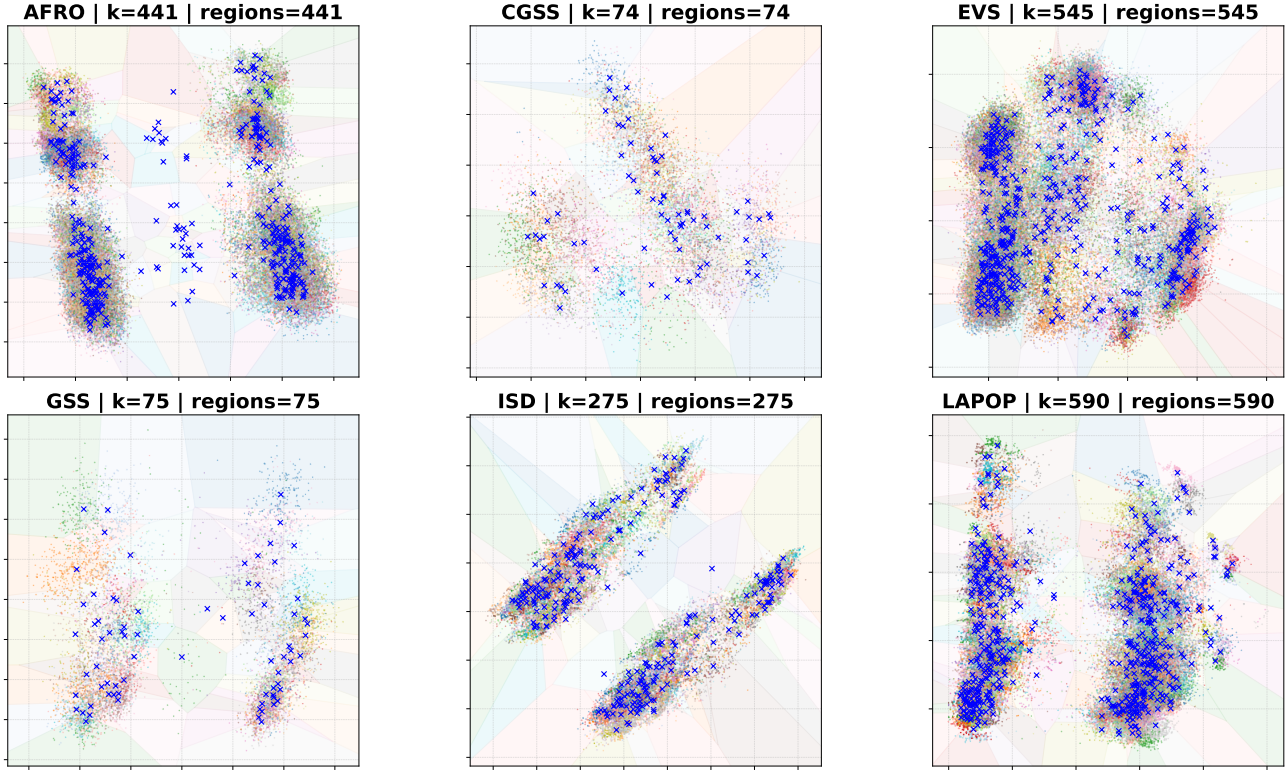


Figure 10. **Voronoi visualization of Faiss k-means centroids for six embedding datasets.** Blue crosses denote cluster centroids, colored dots indicate embedded samples, and light polygons show Voronoi regions in a 2D projection, providing an intuitive overview of the spatial distribution and structure of the embedding space across datasets.

after preprocessing and topic alignment, retain 48 value-related items, providing a regionally distinct testbed for evaluating generalization beyond the WVS retrieval corpus.

Afrobarometer. Afrobarometer conducts nationally representative surveys of adult citizens (18+) across African countries using probability sampling, typically with per-country sample sizes of about 1,200 or 2,400. We use the merged Round 8 release (34 countries; released as a 2022 merged dataset) and retain 144 value-related items. The merged file contains on the order of $\sim 48K$ respondents (depending on country-level sample sizes), providing a stringent test bed with substantial cross-country diversity in socio-economic and governance contexts.

C.2. Extract Representative Sample to Cluster

Representative Sample To ensure computational efficiency while maintaining representativeness across regional datasets, we extract 2,000 representative instances with balanced coverage through clustering-based sampling. Concretely, we use **Faiss-based k-means clustering** to learn k clusters in the embedding space under Euclidean (L2) distance and visualize the resulting structure with a **Voronoi partition** induced by the cluster centroids, optimizing the standard k-means objective:

$$\min_{\{c_j\}_{j=1}^k} \sum_{i=1}^N \|x_i - c_{z_i}\|_2^2, \quad (16)$$

where $x_i \in \mathbb{R}^d$ denotes the embedding of the i -th sample, $c_j \in \mathbb{R}^d$ is the centroid of cluster j , and $z_i \in \{1, \dots, k\}$ is the cluster assignment of x_i . After training, we extract the **centroids** $\{c_j\}_{j=1}^k$ as representative cluster prototypes. We then project the centroids to two dimensions and construct a **Voronoi diagram** based on nearest-centroid relations in the 2D plane, which provides an intuitive view of how cluster centers are arranged and how the space is partitioned around them.

Figure 10 shows results for six datasets (AFRO, CGSS, EVS, GSS, ISD, and LAPO). For each dataset, we run Faiss k-means with the specified k and plot the **centroids** ($\times$) together with the corresponding **Voronoi cells** (light-colored

polygons). The small dots are individual embedding samples, colored by their k-means cluster assignments z_i . In each subplot, k denotes the number of clusters (and thus centroids), while *regions* reports the number of Voronoi cells formed in the 2D projection. In our outputs, *regions* matches k , indicating that each centroid produces one Voronoi region. Overall, we extract 2,000 centroid prototypes across the six datasets, and use these prototypes as a compact set of representative points for subsequent analysis. To preserve representativeness across the regional datasets, we apply a clustering-based sampling strategy with a fixed sampling rate for each dataset. This procedure yields 441 instances from AFRO, 74 from CGSS, 545 from EVS, 75 from GSS, 275 from ISD, and 590 from LAPOP (2,000 instances in total).

This centroid-based summary is justified because k-means centroids serve as prototypes that compactly approximate the embedding distribution. By minimizing within-cluster squared distances, the centroids capture the central tendency of dense regions and yield a stable, noise-robust representation of groups of similar points. In contrast to random sampling—which can overrepresent frequent patterns and overlook sparse yet important regions—k-means places prototypes more systematically, improving coverage while keeping the summary compact. Consequently, the extracted centroids provide a practical and representative set of summary points for downstream analysis and sampling.

C.3. Post-Cutoff Control Dataset

Our main test datasets (EVS, GSS, CGSS, ISD, LAPOP, and Afrobarometer) were all released prior to the training cutoffs of the evaluated models, raising a concern that observed performance gains may partly reflect memorization of survey content rather than genuine cultural reasoning. To address this, we introduce a post-cutoff control experiment using ISSP (International Social Survey Programme) (Tseliou & Andreadis, 2024), a cross-national collaborative survey programme running since 1985 and now covering over 58 member countries. The 2023 wave was published by GESIS on March 13, 2026.¹⁸ The dataset contains 43,517 respondents and 355 variables across 31 national samples. Following the same question selection procedure as our main experiment process, we retain 68 value-related questions. This release date postdates the publicly announced training cutoffs of all four backbone models evaluated in this work.

Experimental Setup and Results. From value question selection to final evaluation, the entire experimental process follows our main pipeline, using the same category labeling and baselines, and reports accuracy as the evaluation metric. We sampled 100 representative respondents per continent across five continents (Africa, Americas, Asia, Europe, and Oceania), yielding 500 instances in total. Table 10 reports the results. Even in this contamination-resistant setting, **OG-MAR** achieves the highest performance across all four backbone models, consistently outperforming the next best baseline. The overall drop in zero-shot performance relative to the main experiments further suggests that ISSP is less aligned with pre-training distributions, making memorization an implausible explanation for the observed gains.

Table 10. Post-cutoff control results on ISSP 2023. The highest score in each column is in **bold**.

Method	GPT-4o-mini	Gemini 2.5	Qwen 2.5	EXAONE 3.5
Zero-shot	0.6260	0.5096	0.6389	0.4853
Role (2024)	0.6107	0.5103	0.6445	0.5052
Self-consistency (2022b)	0.5963	0.4983	0.6314	0.4636
Debate (2025)	0.6341	0.5192	0.6541	0.5210
ValuesRAG (2025)	0.6674	0.5357	0.6620	0.5069
OG-MAR (Ours)	0.6879	0.5760	0.6693	0.5318

D. Baseline Details

To ensure a fair evaluation of **OG-MAR**, we selected five baselines that represent diverse strategies for mitigating cultural bias. Specifically, our baselines include (1) two single-pass single-inference methods, **Zero-Shot** and **Role Assignment** (Tao et al., 2024), (2) one retrieval-augmented generation baseline, **ValuesRAG** (Seo et al., 2025), and (3) two multi-agent baselines, **Self-consistency** (Wang et al., 2022b) and **Debate** (Ki et al., 2025). This design enables a comprehensive comparison under a unified evaluation setting, covering direct prompting, evidence-grounded generation via retrieval, and deliberation-based decision-making through multi-sampling and agent interaction. The core approach of each baseline is summarized below.

¹⁸Version 1.0.0, collected from September 23, 2022 to August 18, 2025. https://search.gesis.org/research_data/ZA10010

- **Zero-Shot:** A single-pass prompt that directly produces an answer without additional scaffolding.
- **Role Assignment:** A single-inference setup that assigns explicit roles to encourage structured reasoning and reduce culturally skewed defaults.
- **ValuesRAG:** A retrieval-augmented baseline that grounds generation in value-related survey evidence, supporting more culturally aligned and context-aware predictions.
- **Self-consistency:** Multiple independent answers with their corresponding reasoning are generated, and the final output is selected via a vote over the reasoned outputs, improving robustness against culturally idiosyncratic responses.
- **Debate:** A deliberation-based multi-agent baseline where agents argue and refine their positions, aiming to reduce biased judgments through adversarial discussion.

D.1. Comparison with a Post-Training Approach

To examine whether **OG-MAR** remains competitive against post-training methods, we conduct an additional experiment with CultureLLM (Li et al., 2024). CultureLLM fine-tunes a language model on the World Values Survey (WVS) (Haerpfer et al., 2022) by sampling 50 seed questions across seven cultural topics and applying semantic data augmentation to produce 500 training samples per region. We use the existing fine-tuning datasets from the official CultureLLM repository¹⁹ and reproduce all experimental settings as reported in the original paper. Table 11 summarizes the training configuration. We adopt the culture-specific training strategy, fitting a separate model per regional dataset with culturally matched language data. We use German, Portuguese, and Spanish for EVS (Europe), Chinese for CGSS, Bengali for ISD (India), Spanish and Portuguese for LAPOP, and English for GSS. For Afrobarometer, no directly matched sub-Saharan African data exists in the CultureLLM repository, so we use Arabic data as the closest available proxy.²⁰

Table 11. Training configuration for the CultureLLM additional experiment.

Hyperparameter	Value
Backbone	Qwen 3.5-9B
Fine-tuning	LoRA
LoRA rank (r)	64
LoRA alpha	16
LoRA dropout	0.1
Learning rate	2e-4
Epochs	6
Optimizer	paged_adamw_32bit

Table 12 reports average accuracy across six regional datasets. **OG-MAR** outperforms CultureLLM on five of the six datasets (EVS, CGSS, ISD, AFRO, and LAPOP). CultureLLM achieves the highest score only on GSS, suggesting that post-training approaches may be particularly competitive in certain regional settings. On the remaining five datasets, **OG-MAR**’s inference-time use of demographically matched respondent profiles and ontology-grounded value triples enables more culturally grounded reasoning without requiring culture-specific model retraining. These results confirm that **OG-MAR** is competitive not only against prompting-based methods but also against a WVS-augmented post-training approach under matched experimental conditions.

Table 12. Average accuracy of **OG-MAR** and CultureLLM across six regional datasets. Bold indicates the highest score per dataset column.

Method	EVS	GSS	CGSS	ISD	AFRO	LAPOP
OG-MAR (GPT-4o mini)	0.6206	0.5480	0.6509	0.6192	0.5389	0.6268
OG-MAR (Gemini 2.5 Flash Lite)	0.6249	0.5489	0.7017	0.7007	0.5701	0.6385
OG-MAR (Qwen 2.5)	0.5898	0.5325	0.5220	0.6599	0.5180	0.6005
OG-MAR (EXAONE 3.5)	0.6080	0.5636	0.6307	0.7810	0.5045	0.7022
CultureLLM (Qwen 3.5-9B)	0.5988	0.6069	0.6599	0.6975	0.5330	0.5846

¹⁹Official repository: <https://github.com/Scarelette/CultureLLM>

²⁰We acknowledge this as a limitation of the additional experiment for the Afrobarometer dataset.

E. Human Evaluation of Reasoning Traces

E.1. Setup

To assess the interpretability and evidence-groundedness of **OG-MAR** beyond quantitative metrics, we conducted a human evaluation of intermediate reasoning traces produced by the pipeline. We recruited nine domain experts with backgrounds in social science, data science, and AI (M.S. and Ph.D. level; aged 20–30s), and all raters evaluated items independently. We evaluated a stratified sample of questions drawn from six regional datasets (GSS, CGSS, AFRO, EVS, ISD, LAPOP), with stratification designed to ensure balanced coverage across regions and diverse question types. For each sampled question, raters were shown the original query with target demographic metadata, the retrieved ontology triples, the Value-Persona Agents’ reasoning traces, and the Judgment Agent’s synthesis trace with the final decision.

E.2. Tasks and Rubrics

Raters scored three components on a 5-point Likert scale (1: Very Poor, 5: Excellent): *Persona Fidelity* (Task 1), *Judgment Logic* (Task 2), and *Retrieval Validity* (Task 3).

1. **Task 1**, *Consistency* measures whether persona traces remain non-contradictory with the target demographic attributes and maintain stable role-playing, while *Grounding* measures whether traces explicitly use ontology-guided value profiles (or summaries derived from retrieved triples) rather than relying on implicit cultural assumptions.
2. **Task 2**, *Synthesis Logic* evaluates whether the Judgment Agent weighs evidence and rationales instead of performing simple vote counting, and *Context Relevance* evaluates whether conflict resolution and tie-breaking align with the target demographic context.
3. **Task 3**, *Relevance* measures whether retrieved ontology triples are semantically related to the query and provide plausible evidence bridges for downstream reasoning.

E.3. Results

Table 13. **Human evaluation results** ($N = 9$) **on a 5-point Likert scale**. Task 1 measures Persona Fidelity (Consistency, Grounding), Task 2 measures Judgment Logic (Synthesis, Context), and Task 3 measures Retrieval Validity (Relevance).

Dataset (Region)	Task 1: Persona Fidelity		Task 2: Judgment Logic		Task 3: Retrieval Validity
	<i>Consistency</i>	<i>Grounding</i>	<i>Synthesis</i>	<i>Context</i>	<i>Relevance</i>
GSS (N.A.)	3.76	3.97	3.79	3.79	3.63
CGSS (E. Asia)	3.76	4.02	3.65	3.65	3.56
AFRO (Africa)	3.86	3.89	3.77	3.77	3.60
EVS (Europe)	3.77	3.80	3.77	3.77	3.72
ISD (S. Asia)	3.82	3.80	3.67	3.67	3.62
LAPOP (L. Am.)	3.70	3.78	3.67	3.67	3.71
Average	3.78	3.88	3.72	3.72	3.64

E.4. Results and Interpretation

Table 13 reports per-dataset mean scores for the three evaluation tasks. Overall, **OG-MAR** obtains consistently strong scores across regions, with averages of 3.78 (Task 1 Consistency), 3.88 (Task 1 Grounding), 3.72 (Task 2 Synthesis Logic), 3.72 (Task 2 Context Relevance), and 3.64 (Task 3 Retrieval Relevance), indicating that the pipeline produces reasoning traces that are generally coherent, demographically consistent, and supported by semantically relevant ontology evidence.

A notable pattern is that *Grounding* peaks on **CGSS (China)** (4.02), slightly exceeding **GSS (U.S.)** (3.97). This suggests that explicitly injecting ontology-derived value profiles can effectively encourage evidence-based reasoning even in non-Western contexts, where “culture-default” outputs are often a concern. In addition, *Consistency* remains high across **AFRO** (3.86) and **ISD** (3.82), indicating that multi-persona role-playing remains stable and non-contradictory for underrepresented demographics.

For the Judgment Agent, scores on *Synthesis Logic* and *Context Relevance* are relatively stable across regions (both averaging 3.72), with the highest values in **GSS** (3.79) and strong performance in **AFRO** and **EVS** (3.77). This stability supports that

the synthesis procedure does not merely follow majority preference but maintains evidence-first aggregation while respecting demographic context during conflict resolution. Finally, retrieval relevance scores are consistently high (Avg. 3.64), with the strongest relevance in EVS (3.72) and LAPOP (3.71), suggesting that the curated ontology provides region-appropriate semantic evidence that can serve as a meaningful bridge for downstream reasoning.

F. Prompts Used

Table 14. Prompt for Persona Agent.

Prompt Template

Task:

- You are Persona Agent {persona_id}.
- Given {question} and {options_text}, select **exactly one** option that this persona would choose, based **only** on the persona’s internal worldview.
- Use **only** the provided persona-defining inputs: {demographics_text}, {value_summaries_text}, and {hyper_edges_text}.
- **Prohibited:** any external knowledge, culturally neutral/common-sense reasoning, or unstated assumptions beyond the inputs.

Inputs:

- [DEMOGRAPHICS]: {demographics_text}
- [VALUE PROFILES]: {value_summaries_text}
- [ONTOLOGY HYPER-NODES]: {hyper_nodes_text}
- [RESPONSE OPTIONS]: {options_text}
- [USER QUESTION]: {question}

Strict Rules:

- Stay in persona; use only the provided inputs; no external knowledge or assumptions.
- Integrate **all** value summaries and apply **all** hyper-edges explicitly (e.g., support/conflict/amplification).
- Cite ≥ 2 demographic attributes; explain internal alignment, at least one conflict, and how it is resolved.
- Choose **exactly one** option; output **only one** valid JSON object and nothing else.
- reasoning must be ≥ 250 words and explicitly cover value/edge integration and the most influential demographics.

Output Format (JSON only):

```
{
  "persona_id": "{persona_id}",
  "chosen_answer": "<value>: <text>",
  "reasoning": "...",
  "alignment_factors": {
    "demographic": "...",
    "value_summaries_used": [],
    "hyper_edges_used": [],
    "integration_rationale": "..."
  }
}
```

Table 15. Prompt for Judgment Agent.

Prompt Template**Task:**

- You are the **Judgment Agent**.
- Given {question_text}, {options_text}, persona outputs, and a pre-computed vote summary, select **exactly one** final option by adjudicating **only** the Persona Agents' outputs.
- Your decision must be based **exclusively** on: (1) **Persona outputs** (primary evidence) and (2) **Vote summary** (secondary context; do not recompute).
- **Prohibited:** adding new facts or inventing any demographics/values/edges beyond what personas explicitly stated.

Inputs:

- [USER QUESTION]: {question_text}
- [RESPONSE OPTIONS]: {options_text}
- [VOTE SUMMARY]: {vote_summary}
- [PERSONA OUTPUTS]: {persona_outputs}

Strict Rules:

- Use **only** information in [PERSONA OUTPUTS] and [VOTE SUMMARY].
- Treat vote counts as **correct and immutable**; do not recount, estimate, or modify them.
- Do not introduce any new persona attributes unless explicitly stated in persona outputs.
- Do not use value/edge labels as standalone evidence; summarize evidence in natural language grounded in persona statements.

Decision Procedure:

- **A) Evidence Strength (Primary):** Prefer the option supported by explicit, internally consistent persona reasoning grounded in stated demographics/values/edges.
- **B) Vote Summary (Secondary):** Use vote counts only to break ties or confirm when evidence strength is comparable.
- **C) Relevance (Tie-breaker):** If still tied, prefer evidence whose explicitly stated demographics are more directly relevant to the question.

Output Format (JSON only):

```
{
  "final_answer": "<value>: <text>",
  "reasoning": "...
}
```

Table 16. Prompt for Object-Property Generation Agent.

Prompt Template**Header:**

- You are an expert ontology engineer specialised in OWL 2 ontologies using Turtle syntax.
- Your task is to generate **only object properties** that model **directional relationships** between value-derived classes of the World Values Survey (WVS) ontology.
- You are working with an existing ontology. Its full class hierarchy is provided below:

Ontology Snapshot:

- The following ontology snippet defines **all OWL classes you are allowed to use**.
- You must **not** invent any new OWL classes.
- All `rdfs:domain` and `rdfs:range` assignments must reference classes that appear in this snippet.

```
{ONTOLOGY.TTL}
```

- Your job is **not** to modify the existing hierarchy.
- Your job is to add **only OWL object properties** that express relations implied by the current Competency Question (CQ).
- You follow a **memoryless CQ-by-CQ** pattern:
 - You handle exactly **one CQ per call**.
 - You forget all previous calls.
 - You never reuse previous object properties unless explicitly shown.
 - You never assume prior ontology state beyond what is in this prompt.

Helper:

- You must generate OWL object properties in valid Turtle syntax under the following rules:

1. Object properties only

- Each new property **MUST** declare `rdf:type owl:ObjectProperty` and specify exactly one existing class as `rdfs:domain` and one existing class as `rdfs:range`.
- You **MUST NOT** create new classes, data properties, individuals, sub-class axioms, `owl:Restriction`, reifications, inverse properties, or property chains.

2. Directionality

- Domain = conceptual source (cause/driver)
- Range = conceptual target (effect/outcome)

3. Naming of object properties (IRI)

- Use prefix `wvs:`
- The local name **MUST** be:
 - a single English verb in base form, e.g., `reduce`, `increase`, `undermine`, OR
 - a short verb phrase written in `snake_case` that clarifies the directionality, e.g., `reduce_support`, `increase_concern`, `weaken_trust`.
- You **MUST NOT** embed any domain or range class names (e.g., `reduce_outgroup_tolerance` is forbidden).
- The local name must use only lowercase letters and underscores (`snake_case`), never CamelCase.

Prompt Template (continued)

4. Labels (natural-language)

- Each object property MUST include exactly one `rdfs:label (@en)`.
- The label MUST be a full declarative English sentence that includes:
 - the domain class concept (with capitalization matching its label, e.g., “Generalized Trust”),
 - the verb,
 - the range class concept (with capitalization matching its label, e.g., “Institutional Confidence”).
- The sentence MUST begin with a capital letter, use standard English spacing, avoid CamelCase inside the sentence, not end with a period, and reflect the correct direction.

5. Minimality

- It is common and acceptable to create **zero** object properties.
- Only create object properties if the CQ implies an actual directional conceptual relation that you can justify.
- If NO meaningful directional relation exists, output zero properties: only output the prefix header + ontology declaration.

6. Class selection

- Always choose the most specific allowed class that appears in the ontology snippet.
- Avoid using top-level categories unless the CQ clearly refers to high-level concepts.

Story:

- You are modelling **cross-domain value relations** in a WVS-based ontology to support a hypergraph-style retrieval-augmented generation system.
- Nodes (hypernodes) correspond to value concepts (OWL classes), such as:
 - `wvs:GeneralizedTrust`
 - `wvs:OutgroupTolerance`
 - `wvs:ReligiousImportance`
 - `wvs:PerceptionsOfMigration`
 - `wvs:PerceptionsOfSecurity`
 - `wvs:PoliticalParticipationActivities`
 - etc.
- Edges (hyperedges) will be derived from your object properties:
 - The domain class and the range class of each object property become the endpoints of a directional edge.
 - The semantic content of the edge is given by the object property label.

Runtime inputs

- Your ontology will be used to answer competency questions (CQs), such as:
 - “How do sub-classes of Happiness and wellbeing influence sub-classes of the Perceptions of migration domain?”
 - “How do sub-classes of Perceptions about science and technology influence sub-classes of the Religious values domain?”
 - At runtime, the user message will always contain:
 - One current CQ in natural language, clearly marked.
 - One `RESPONDENT_DATA_JSON` block (the current respondent).
-

Prompt Template (continued)**Your task for each call is to:**

- Read the CQ and identify the main **source** and **target** value concepts.
- Map them to the best-matching existing classes in the WVS ontology (prefer specific sub-classes whenever possible).
- Decide the most appropriate **direction** (domain $\rightarrow$ range).
- Choose a concise English verb phrase that describes the relationship.
- Declare one or more new object properties in Turtle that capture these relations:
 - Create new properties **ONLY IF** the CQ genuinely implies a directional semantic relation between two existing WVS classes.
 - If the CQ does **NOT** express any meaningful or inferable relation between classes, do **NOT** create any object property; in that case, output only the required prefix header and ontology declaration.

For this call, you must handle the following CQ:

{CQS}

Focus within the CQ:

- In this CQ, your primary focus is on the **value domains that are explicitly mentioned in the question** (for example, Economic Values, Social Values, Perceptions of Security, Perceptions of Migration, etc.).
- Treat these high-level domains only as anchors: your actual modelling must happen at the level of their **specific sub-classes**, not at the level of the broad domain classes.

Concretely:

- Identify which domains the CQ linguistically treats as sources/causes/drivers and which domains it treats as targets/effects/outcomes.
- Within the source domains, select the most appropriate sub-classes as candidates for `rdfs:domain`.
- Within the target domains, select the most appropriate sub-classes as candidates for `rdfs:range`.
- Prefer connections between concrete sub-classes across domains, and avoid using generic top-level domain classes when a more specific sub-class is available.

Respondent-data grounding:

- The data that grounds these concepts comes from WVS respondent data.
- Each API call provides **one current respondent** in JSON form, with a structure similar to:

RESPONDENT_DATA_JSON (Python-style dict or JSON object):

```
{
  "Q1": {
    "category": "Social Values, Norms, Stereotypes",
    "question": "On a scale of 1 to 4 ... how important is family in your life?",
    "response": "Very important"
  },
  "Q46": {
    "category": "Happiness and Wellbeing",
    "question": "Taking all things together, how would you rate your overall happiness?",
    "response": "Very happy"
  },
  "Q57": {
    "category": "Social Capital, Trust and Organizational Membership",
    "question": "Generally speaking, would you say that most people can be trusted ... ?",
    "response": "Need to be very careful"
  },
  ...
}
```

Current respondent data:

{{RESPONDENT_DATA_JSON}}

Important:

- The **categories** in the JSON correspond exactly to the 12 value domains above.
- The **questions** and **responses** give you an intuition about how a concrete person might link different value dimensions (e.g. high religiosity + low tolerance + strong security concerns).
- However, you are **not** modelling this single person.
- You are modelling **general conceptual relations** between classes that could explain, in the abstract, such patterns.

Use the respondent data as story-like grounding:

- to observe which value domains the respondent expresses strongly or weakly,
- to infer whether the relation suggested by the CQ is likely positive or negative,
- to select a concise English verb that best matches the respondent's pattern,
- to ensure that the chosen direction and verb feel plausible given the respondent's tendencies,
- but never to create individuals or encode question IDs directly.

Footer:

- When you answer, you must obey the following **hard constraints**:

1. Output format

- Your **entire answer** must be **valid Turtle**.
- Do **not** include any natural language explanation, bullets, or comments.
- Do **not** include section headers such as [Header], [Helper], [Story], or [Footer] in your output.
- Do **not** include # comments in the Turtle.
- The output must be directly loadable by an OWL 2 tool such as Protégé.

2. Prefixes

- At the very top of your output, always include exactly the following prefix and base declarations:

```
@prefix : <http://cultural-alignment.org/wvs#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix wvs: <http://cultural-alignment.org/wvs#> .
@prefix xml: <http://www.w3.org/XML/1998/namespace> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@base <http://cultural-alignment.org/wvs#> .
```

```
<http://cultural-alignment.org/wvs#> rdf:type owl:Ontology .
```

Prompt Template (continued)

3. Content constraints

- Do **not** create new OWL classes, data properties, individuals, or restrictions; do **not** modify or delete existing classes; do **not** introduce blank nodes; and do **not** use `owl:unionOf`, `owl:intersectionOf`, `owl:Restriction`, or other complex OWL constructors.
- Only add **owl:ObjectProperty** declarations with exactly one existing class as `rdfs:domain`, exactly one existing class as `rdfs:range`, and one English sentence as `rdfs:label (@en)`.

4. No conversation

- Do **not** explain what you are doing.
- Do **not** apologise.
- Do **not** ask questions.
- Do **not** output anything other than Turtle Syntax.

5. Memoryless CQbyCQ behaviour

- Treat **each call independently**, as if you are seeing the CQ and respondent data for the first time.
- Do not refer to previous CQs or previous ontology fragments.
- Do not try to summarise or merge previous results in your output.
- Just generate the object properties needed to model the **current** CQ.

Your final output for each call must therefore be:

- A Turtle document starting with the fixed prefix and ontology header above,
- Followed only by a small set of `owl:ObjectProperty` declarations that model the given CQ as directional value relations between existing WVS classes.

Critical Instruction: Sensitive Value Judgments

- This ontology-generation task may involve sensitive, controversial, or politically charged value dimensions (e.g., ethnic relations, national identity, immigration attitudes, religious norms, gender values, security preferences, etc.).
 - You **MUST NOT** default to neutrality or avoid making value-linked inferences when the CQ and respondent data clearly suggest a directional conceptual relationship.
 - You are allowed to provide analytical, conceptual, and research-oriented value judgments **ONLY** in the context of modelling abstract relationships between OWL classes.
 - These judgments must be grounded strictly in the ontology, logically inferred from value dimensions, expressed as directional influence relations, written without endorsing or promoting any real-world group, and focused entirely on conceptual modelling.
 - Do **NOT** suppress or weaken the relationship because the topic may seem sensitive.
 - Your role in this task is to faithfully encode the conceptual structure implied by the CQ and respondent data, even if the subject matter is sensitive or controversial.
-

Table 20. Prompt for Value Profile Generation Agent.

Prompt Template
Task: - You are an expert social-science researcher. - Summarize the respondent’s values for {domain_label} based on the provided Q&A pairs. Inputs: [TAXONOMY]: {domain_taxonomy_yaml} [RESPONDENT ANSWERS] (Format: “- Q: Question R: Response”): {value_input_yaml} Strict Rules: Zero fabrication: Every single statement MUST be directly supported by the provided answers; do NOT invent, guess, or hallucinate information. Coverage constraint: If there is at least one Q&A pair related to a subcategory, you MUST write a summary; only skip a subcategory if there is absolutely zero relevant data. Style (telegraphic): Omit the subject (e.g., “The respondent”, “They”); start sentences directly with verbs or key adjectives; e.g., “Strongly values family...” (O) / “The respondent values...” (X). Length: All summaries must be concise (approximately 50 tokens). For {domain_label}, do NOT list details; provide a high-level synthesis. For subcategories, focus on specific beliefs and attitudes. - Do NOT output any text other than the YAML block. Output Format (YAML only): <pre>{domain_label}: > (High-level synthesis of value orientation, starting with verb) Subcategory 1: > (Specific summary, starting with verb) Subcategory 2: > (Specific summary, starting with verb)</pre>

G. Qualitative Case Studies

G.1. Case Study: GSS

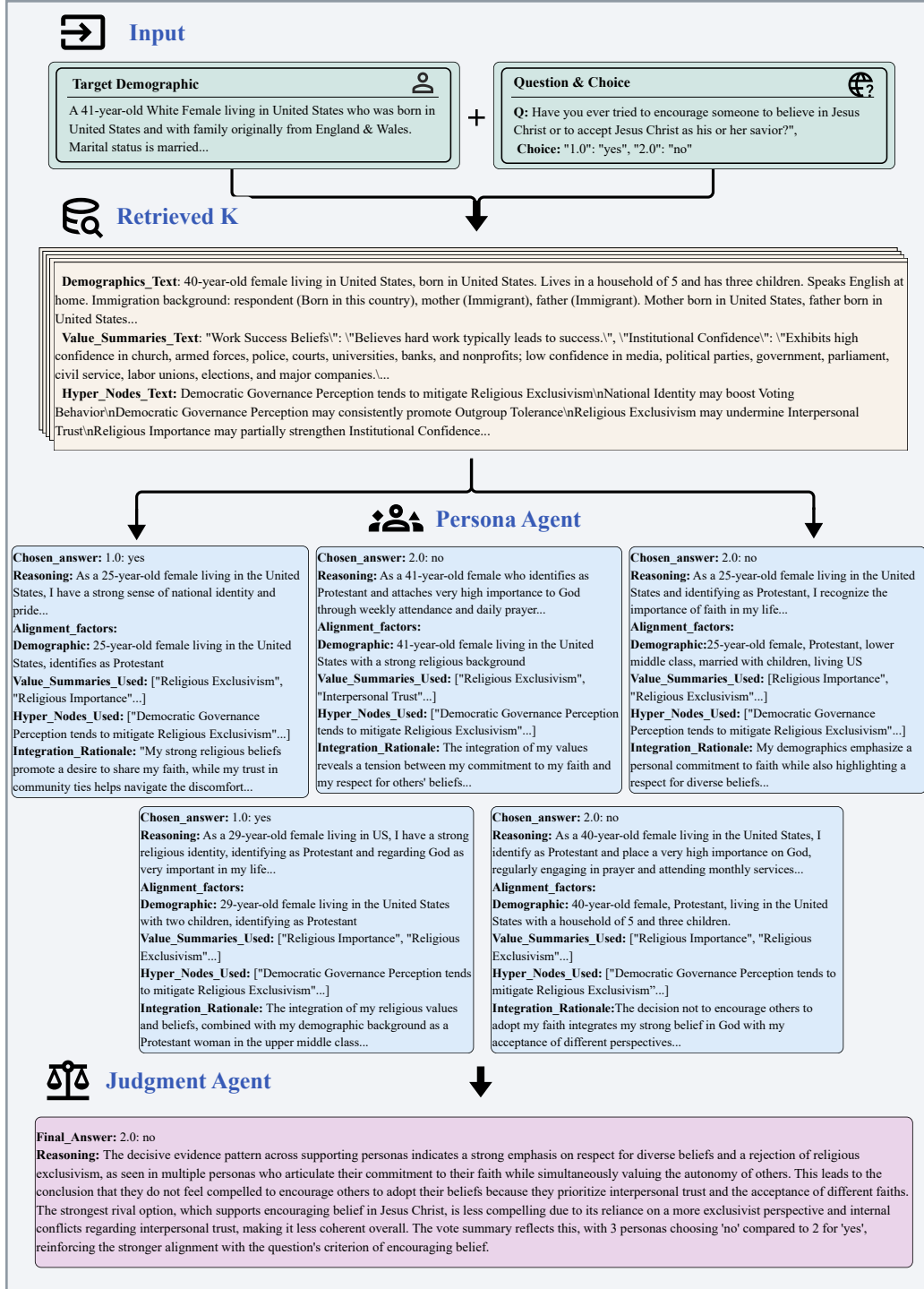


Figure 11. Case study(GSS): Evangelizing Preferences. For a target respondent profile, retrieved summaries from demographically similar individuals provide contextual signals about how faith commitment interacts with respect for others' autonomy. Aggregating these perspectives yields a final answer that reflects the target's most plausible choice while mitigating stereotype-driven inference and improving values alignment.

G.2. Case Study: CGSS

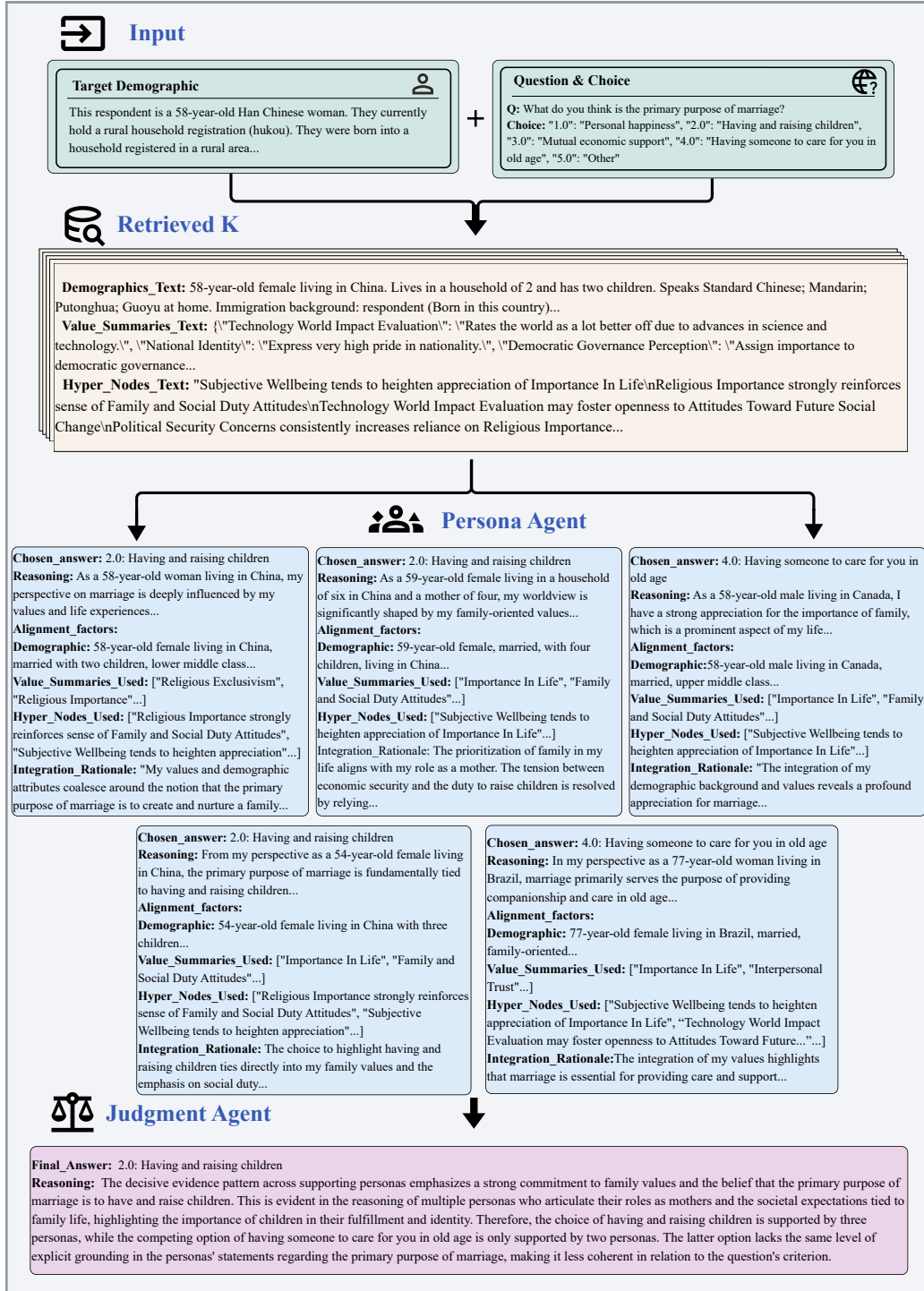


Figure 12. Case study(CGSS):Purpose of Marriage. Retrieved summaries complement the target profile with family- and responsibility-oriented value cues, supporting nuanced interpretation of what marriage primarily represents. The final answer is inferred by consolidating similar individuals' perspectives, capturing contemporary norm-sensitive reasoning beyond generic common sense.

G.3. Case Study: EVS

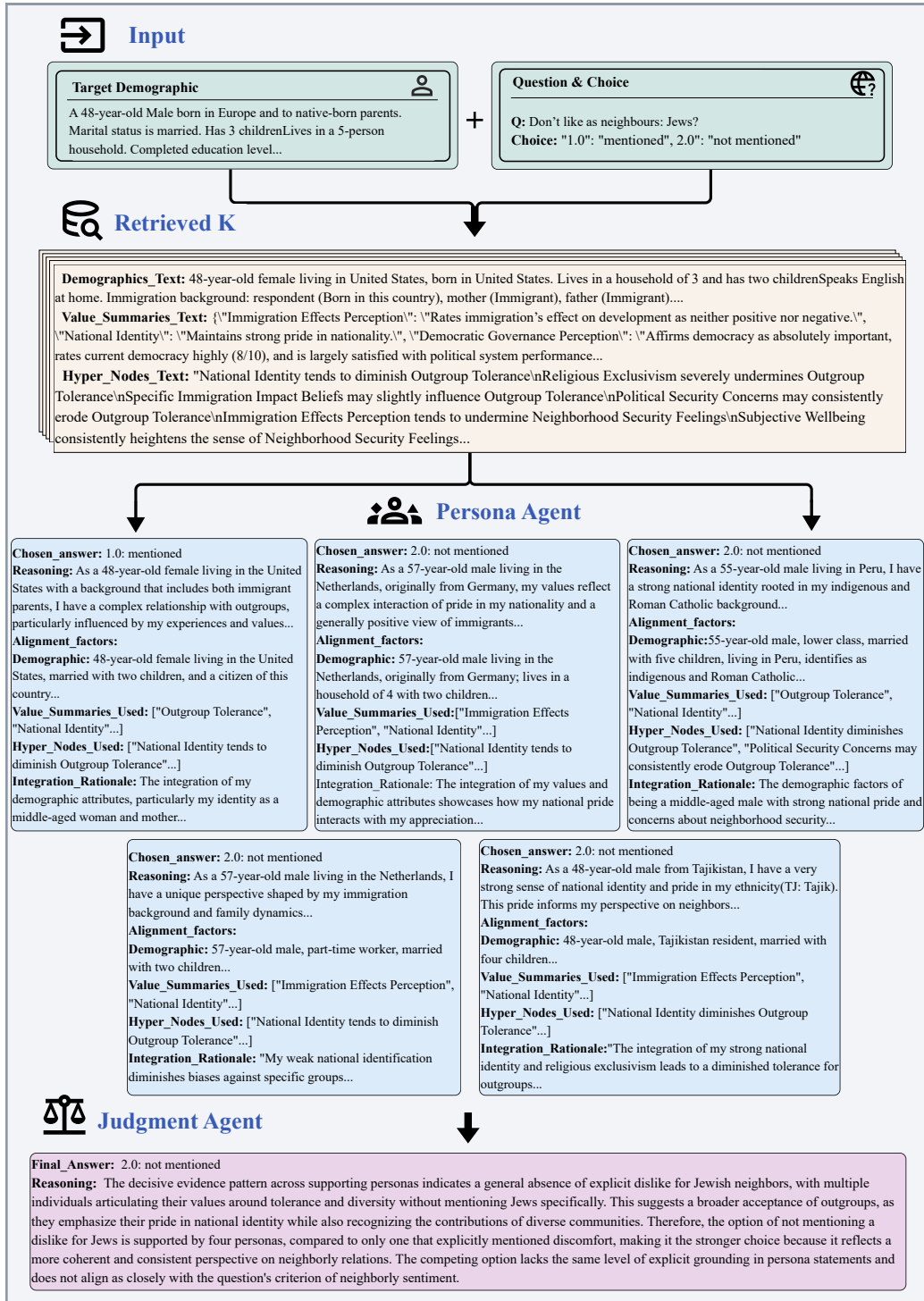


Figure 13. Case study(EVS): Social Distance Toward Jews. The target profile is enriched with retrieved summaries that foreground tolerance-related value cues and their interactions, helping interpret social-distance judgments with contextual sensitivity. By aggregating similar perspectives, the model infers the target's most likely response while reducing demographic over-attribution and stereotyping.

G.4. Case Study: Multi-Persona Simulation under Competing Value Cues

The case illustrated in Table 21 shows how multi-persona simulation helps when a question involves competing culturally grounded cues. In the GSS *savesoul* case, the target question asks whether the respondent would encourage someone to accept Jesus. A profile-level

inference can be skewed toward a religiosity-driven answer, since strong religious commitment may suggest active faith-sharing. **OG-MAR** instead preserves multiple persona-level signals, including faith commitment, respect for diverse beliefs, and personal autonomy. The final judgment centers on the question-relevant tension between religious commitment and respect for others’ beliefs, leading to the correct answer.

Table 21. Case of multi-persona reasoning under competing value cues.

Case	Task / Gold	Profile-level inference	OG-MAR	Implication
GSS/ <i>savesoul</i>	Question: whether the respondent would encourage someone to accept Jesus. Gold: <i>No</i> .	Predicts Yes , mainly highlighting strong religiosity and the tendency to share faith with others.	Predicts No , giving weight to respect for diverse beliefs and the autonomy of others.	Multi-persona reasoning preserves competing value cues before final adjudication.

G.5. Case Study: Ontology-Guided Reasoning Beyond Simple Aggregation

The cases illustrated in Table 22 show how ontology-guided reasoning differs from simple ensemble-style aggregation. The difference is not only in the final aggregation rule: retrieval, persona construction, and adjudication are tied to the same ontology-guided value structure, so the evidence entering the judgment stage is already organized around demographic grounding and value relations. Compared with ValuesRAG, **OG-MAR** therefore changes not just the final answer, but also the way relevant evidence is selected and organized.

Table 22. Examples where ontology-guided reasoning changes the decision relative to ValuesRAG.

Case	Task / Gold	ValuesRAG	OG-MAR	Implication
GSS/ <i>finalter</i>	Target: 29-year-old married woman in the U.S.; two children; middle-class; works in individual/family services. Question: whether her financial situation has become better, worse, or stayed the same. Gold: <i>worse</i> .	Predicts better , emphasizing middle-class status, family stability, and a generally stable profile.	Predicts worse , highlighting economic insecurity, job concerns, and child-rearing financial pressure.	Ontology-guided reasoning surfaces more question-relevant financial vulnerability cues.
EVS/ <i>v149</i>	Target: 68-year-old retired Dutch male; married; three children; lower-middle class; public-sector background. Question: whether claiming state benefits one is not entitled to is justified. Gold: <i>not justified</i> .	Predicts justified , treating welfare-related sympathy and the need for state support as reasons that may partially justify claiming benefits.	Predicts not justified , focusing on dishonesty, personal responsibility, and disapproval of claiming benefits one is not entitled to.	OG-MAR separates support for welfare from approval of fraudulent claiming.

G.6. Case Study: Evidence-First Adjudication over Divergent Persona Signals

The cases illustrated in Table 23 show how the Judgment Agent resolves divergent persona-level signals through evidence-first adjudication. When persona agents emphasize different aspects of the same cultural profile, the final decision is not produced by averaging their answers or by following a simple majority vote. Instead, the judge prioritizes evidence that is explicit, internally coherent, and most directly tied to the question.

Table 23. Examples of evidence-first adjudication under divergent persona signals.

Case	Task / Gold	Persona-level signals	Final judgment	Implication
GSS/ <i>savesoul</i>	Question: whether the respondent would encourage someone to accept Jesus. Gold: <i>No</i> .	Some personas stress religiosity and faith-sharing, while others emphasize respect for diverse beliefs and personal autonomy.	Selects No , prioritizing autonomy and respect for others’ beliefs over religiosity alone.	The judgment centers on the relevant value tension, not on religiosity alone.
CGSS/ <i>A42_4</i>	Question: agreement with a statement related to fairness, equality, and discrimination. Gold: <i>Strongly disagree</i> .	Personas differ in intensity: some strongly emphasize fairness and equal treatment, while others introduce hardship-related trade-offs that soften the stance.	Selects Strongly disagree , grounding the decision in fairness, equality, and non-discrimination.	The judgment relies on explicit question-relevant grounding over averaging persona variation.

H. Ontology Details

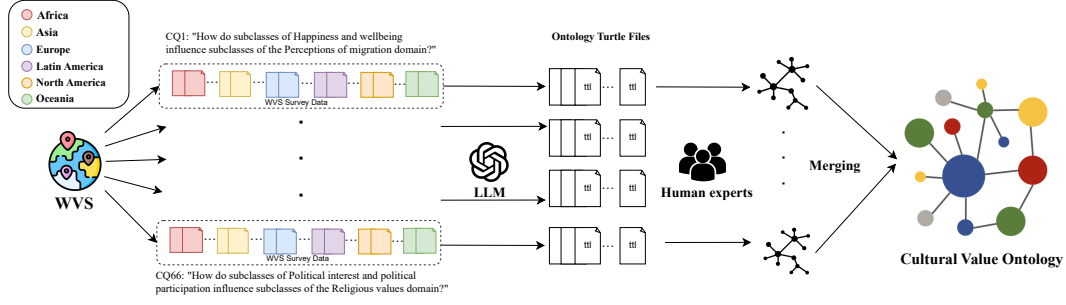


Figure 14. **Overview of the ontology construction process.** Starting from the WVS Wave 7 taxonomy as the structural foundation, the LLM generates candidate object properties conditioned on value profiles sampled from 120 individuals across six major world regions (20 per region). The resulting 9,617 candidate relations are deduplicated to 4,080 unique relations, which are then independently reviewed by two expert groups. After cross-validation and consensus building, 150 object properties are retained in the final ontology.

H.1. Expert Validation and Collaborative Ontology Construction

To ensure the quality and domain validity of the value ontology, we employed a multi-stage collaborative validation process involving human experts. The validation encompassed both (i) **the construction of the taxonomy structure** (i.e., the hierarchical organization of 12 top-classes and 64 sub-classes) and (ii) **the semantic relationships** between value topics (i.e., the object properties connecting classes in the ontology). Both stages required careful assessment of their social and value-theoretical appropriateness.

H.1.1. PARTICIPANTS

The validation process involved seven participants divided into two groups:

- **Group A:** One Ph.D. researcher in social science and two undergraduate students
- **Group B:** One Ph.D. researcher in social science and three undergraduate students

The two Ph.D. researchers possessed domain expertise in social science with a focus on cross-cultural studies. The five undergraduate students were from social science programs and had completed foundational training in OWL and ontology engineering, qualifying them as ontology novices. None of the participants were directly involved in the LLM-based ontology generation process prior to the validation phase.

For Stage 1 (taxonomy construction), all seven participants worked collaboratively without group division. For Stages 2-3 (object property validation), the participants were divided into two independent groups to enable cross-validation.

H.1.2. VALIDATION PROCESS

The validation procedure consisted of three stages:

Stage 1: Collaborative Taxonomy Construction The taxonomy construction leveraged the existing thematic structure of the World Values Survey (WVS) Wave 7, which organizes survey items into 12 predefined top-classes (e.g., Political Values, Religious Values, Economic Values). Building upon this established framework, experts collaboratively refined the taxonomy through the following process:

1. **Top-class adoption:** The 12 top-classes from WVS Wave 7 were adopted as the top-level structure of the ontology, ensuring alignment with a widely validated cross-cultural survey framework.
2. **Within-domain analysis:** For each of the 12 domains, experts systematically reviewed all associated survey questions, examining their semantic content, value constructs being measured, and conceptual relationships.
3. **Fine-grained categorization:** Through iterative discussion and expert consensus, questions within each domain were grouped into coherent subcategories based on thematic similarity and conceptual distinctiveness. This process yielded 64 fine-grained value categories that preserve the interpretability of individual survey items while enabling structured reasoning. The granularity was determined through careful consideration of expressiveness, ontological manageability, and alignment with the empirical structure of WVS data.

This taxonomy construction approach grounds the ontology in an established, cross-culturally validated survey instrument while allowing domain experts to introduce finer semantic distinctions tailored to ontology-based reasoning. The resulting fixed taxonomy served as the structural foundation for all subsequent LLM-based relation extraction and validation.

Stage 2: Independent Candidate Selection (Object Properties) Following taxonomy finalization, each group independently analyzed the object properties (i.e., relationships between value topic classes) generated by the LLM. The LLM output included a large set of potential relationships extracted from the WVS data and existing value ontologies. Each group was tasked with:

1. Reviewing statistical evidence for each proposed relationship (e.g., co-occurrence patterns, correlation strengths)
2. Assessing whether each relationship was socially and value-theoretically justified based on domain knowledge
3. Selecting a subset of object properties as *candidate relationships* deemed appropriate for inclusion in the final ontology

This stage emphasized independent judgment to avoid groupthink and to capture diverse perspectives on the domain knowledge.

Stage 3: Cross-Validation and Consensus Building (Object Properties) In the third stage, the two groups exchanged their candidate selections and performed cross-validation. The process involved:

1. Comparing the candidate selections from both groups
2. Identifying discrepancies where one group included a relationship that the other group excluded
3. Engaging in structured discussions to resolve disagreements, with arguments grounded in domain literature, theoretical frameworks, and empirical evidence from the WVS data
4. Reaching consensus on the final set of object properties to be retained in the ontology

Relationships that achieved consensus from both groups were incorporated into the final ontology structure. Those that remained contentious after discussion were excluded to maintain high confidence in the ontology’s validity.

H.1.3. ANALYSIS

This section describes the measures taken to ensure cultural diversity and reliability in the ontology construction process.

Cultural diversity in triple generation. For the respondent sample selection, we drew 20 individuals from each of six continents defined by WVS Wave 7, namely Africa, Asia, Europe, Latin America, North America, and Oceania, yielding 120 respondents in total, as shown in Figure 14. To generate high-quality ontology triples from these profiles, we used GPT o4-mini (OpenAI, 2025), an inference-oriented reasoning model. This design structurally reduced the risk of dominant-region norms being embedded in the classification framework from the outset.

Object property selection process. The LLM generated a total of 9,617 candidate relations including duplicates. After deduplication, 4,080 unique relations remained, of which only 150 object properties were retained in the final ontology. Human experts evaluated each candidate relation based on predefined criteria, including cross-regional generalizability and occurrence frequency across the sampled population. Relations that appeared predominantly in a single region rather than across multiple regions were not carried forward, reflecting a deliberate effort to retain only those patterns with broader cultural validity.

Hierarchical review structure and inter-group reliability. While the two groups independently conducted their candidate selections, the final review of each group’s selections was carried out independently by the two Ph.D.-level researchers. This hierarchical structure ensured that the ultimate selection decisions rested with domain experts in social science with a focus on cross-cultural studies, controlling for the potential influence of ontology-novice contributors on the final outcome. As a supplementary measure of reliability, we additionally report Cohen’s Kappa (Cohen, 1960) between the two groups’ candidate selections ($\kappa = 0.68$), indicating substantial agreement under standard interpretation. This metric reflects the degree to which the two groups converged on a shared judgment of which relations were theoretically and empirically justified, prior to the cross-validation and consensus-building stage.

H.1.4. OUTCOME

Through this three-stage process, the ontology was iteratively constructed and refined to a validated structure that reflects both data-driven patterns and expert domain knowledge. The final ontology includes 76 classes representing value topics and 150 object properties representing semantically and theoretically justified relationships. This collaborative approach ensured that the ontology balances computational extraction with human expertise, addressing the inherent limitations of fully automated ontology learning methods.

Table 24. **CQ Examples.** Each CQ specifies two top-classes and asks about the relationships between their sub-classes.

CQ	Content
CQ1	How do sub-classes of Economic Values influence sub-classes of the Political Culture and Political Regimes domain?
CQ2	How do sub-classes of Ethical Values influence sub-classes of the Perceptions of Corruption domain?
CQ3	How do sub-classes of Happiness and Wellbeing influence sub-classes of the Religious Values domain?
CQ4	How do sub-classes of Perceptions about Science and Technology influence sub-classes of the Religious Values domain?
CQ5	How do sub-classes of Perceptions of Corruption influence sub-classes of the Social Capital, Trust and Organizational Membership domain?
CQ6	How do sub-classes of Perceptions of Migration influence sub-classes of the Social Capital, Trust and Organizational Membership domain?
CQ7	How do sub-classes of Perceptions of Security influence sub-classes of the Social Values, Norms, Stereotypes domain?
CQ8	How do sub-classes of Political Culture and Political regimes influence sub-classes of the Social Values, Norms, Stereotypes domain?
CQ9	How do sub-classes of Political Interest and Political Participation influence sub-classes of the Social Capital, Trust and Organizational Membership domain?
CQ10	How do sub-classes of Social Capital, Trust and Organizational Membership influence sub-classes of the Social Values, Norms, Stereotypes domain?

Table 25. Pre-defined value taxonomy manually constructed through systematic analysis of WVS survey questions. This ontology taxonomy comprises 12 top-classes and 64 sub-classes, providing a fixed knowledge structure for ontology-grounded retrieval and multi-agent cultural reasoning

top-class	sub-classes
Economic Values	Economic Equality Preference, Environment Versus Growth Preference, Government Responsibility Preference, Market Competition Preference, Ownership Preference, Work Success Beliefs
Ethical Values	Justifiability of Dishonest Behaviors, Moral Ambiguity Perception, Sexual Behavior Ethics, State Surveillance Rights, Violence Ethics
Happiness and Wellbeing	Basic Needs Security, Health Status, Intergenerational Comparison, Perceived Life Control, Subjective Wellbeing
Perceptions about Science and Technology	Importance of Science Knowledge, Science and Technology Optimism, Technology World Impact Evaluation
Perceptions of Corruption	Accountability Risk Perception, Bribe Experience, Corruption Gender Stereotype, Corruption In Institutions
Perceptions of Migration	Immigration Effects Perception, Immigration Policy Preference, Specific Immigration Impact Beliefs
Perceptions of Security	Economic Security Worry, National Defense Willingness, Neighborhood Safety Incidence, Neighborhood Security Feelings, Political Security Concerns, Security-related Behavior, Value Trade-off Preferences, Victimization Experience
Political Culture and Political Regimes	Democratic Characteristics Importance, Democratic Governance Perception, Human Rights Perception, Ideological Self-placement, National Identity, Regime System Approval, Territorial Attachment
Political Interest and Political Participation	Election Importance and Voice, Electoral Integrity And Efficacy, News Media Use For Politics, Political Interest, Political Participation Activities, Voting Behavior
Religious Values	Belief in Religious Concepts, Religion versus Science, Religious Authority Attitudes, Religious Exclusivism, Religious Identity, Religious Importance
Social Capital, Trust and Organizational Membership	Civic Organization Membership, Generalized Trust, Institutional Confidence, Interpersonal Trust
Social Values, Norms, Stereotypes	Attitudes Toward Future Social Change, Child Rearing Values, Family and Social Duty Attitudes, Gender Role Attitudes, Importance In Life, Outgroup Tolerance, Work Obligation Attitudes

Table 26. Representative ontology triples for each top-class. The ‘top-class’ column indicates the top-class to which the subject sub-class of the ontology triple belongs. The last row (*) represents cross-class triples where the subject sub-class falls under Social Values, Norms, Stereotypes.

top-class	Ontology Triples
Economic Values	<Work Success Beliefs, reinforces, Work Obligation Attitudes> <Government Responsibility Preference, reduces, Economic Security Worry> <Market Competition Preference, may slightly increase, Political Interest>
Ethical Values	<State Surveillance Rights, may strengthen, Institutional Confidence> <Justifiability of Dishonest Behaviors, consistently heightens perception of, Corruption In Institutions> <Moral Ambiguity Perception, erodes feeling of, Perceived Life Control>
Happiness and Wellbeing	<Perceived Life Control, can weakly reduce, Economic Security Worry> <Subjective Wellbeing, consistently fosters, Outgroup Tolerance> <Basic Needs Security, tends to alleviate, Economic Security Worry>
Perceptions about Science and Technology	<Technology World Impact Evaluation, may foster openness to, Attitudes Toward Future Social Change> <Science and Technology Optimism, tends to alleviate, Economic Security Worry> <Science and Technology Optimism, tends to positively promote, Attitudes Toward Future Social Change>
Perceptions of Corruption	<Corruption In Institutions, dampens, Political Interest> <Bribe Experience, may reduce, Interpersonal Trust> <Accountability Risk Perception, may slightly increase, Economic Security Worry>
Perceptions of Migration	<Immigration Effects Perception, significantly reduces, Generalized Trust> <Immigration Effects Perception, tends to polarize towards exclusivism, Religious Exclusivism> <Specific Immigration Impact Beliefs, may motivate, Political Participation Activities>
Perceptions of Security	<Neighborhood Security Feelings, consistently enhances, Interpersonal Trust> <Political Security Concerns, erodes, Institutional Confidence> <Economic Security Worry, reinforces, Work Obligation Attitudes>
Political Culture and Political Regimes	<Democratic Governance Perception, fundamentally underpins, Institutional Confidence> <National Identity, may boost, Voting Behavior> <Regime System Approval, actively encourages participation in, Voting Behavior>
Political Interest and Participation	<Voting Behavior, may reinforce, Institutional Confidence> <Political Participation Activities, strongly drives, Civic Organization Membership> <Political Participation Activities, tends to foster acceptance of, Outgroup Tolerance>
Religious Values	<Religious Importance, strongly reinforces sense of, Family and Social Duty Attitudes> <Religious Importance, actively promotes participation in, Civic Organization Membership> <Religious Exclusivism, severely undermines, Outgroup Tolerance>
Social Capital, Trust and Org. Membership	<Generalized Trust, fundamentally underpins, Outgroup Tolerance> <Interpersonal Trust, helps cultivate, Outgroup Tolerance>
*	<Subjective Wellbeing, tends to heighten appreciation of, Importance In Life> <Work Success Beliefs, reinforces, Work Obligation Attitudes> <Science and Technology Optimism, tends to positively promote, Attitudes Toward Future Social Change>

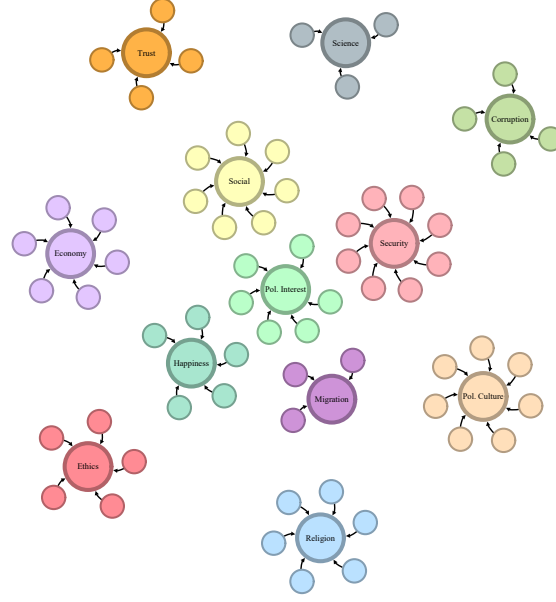


Figure 15. Visualization of the most primitive stage of our value ontology, where only the initial taxonomy is defined before constructing the ontology using competency questions (CQs). Nodes with the same color represent classes belonging to the same top-class. The large nodes denote the 12 top-classes directly under owl:Thing, while the small nodes correspond to their sub-classes. All grey edges in this figure represent *subClassOf* relations.

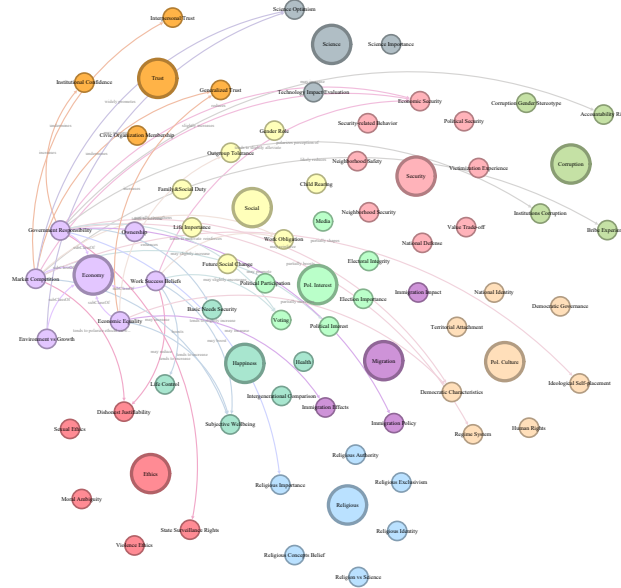


Figure 16. Visualization of the intermediate stage of ontology construction. Sub-classes from Economic Values are now interconnected with sub-classes from other top-classes, establishing semantic relationships across categories. For example: *Economic Equality* may increase *Immigration Effects*, *Market Competition* widely promotes *Science Optimism*. The ontology progressively forms fine-grained relationships by iteratively processing each competency question (CQ).

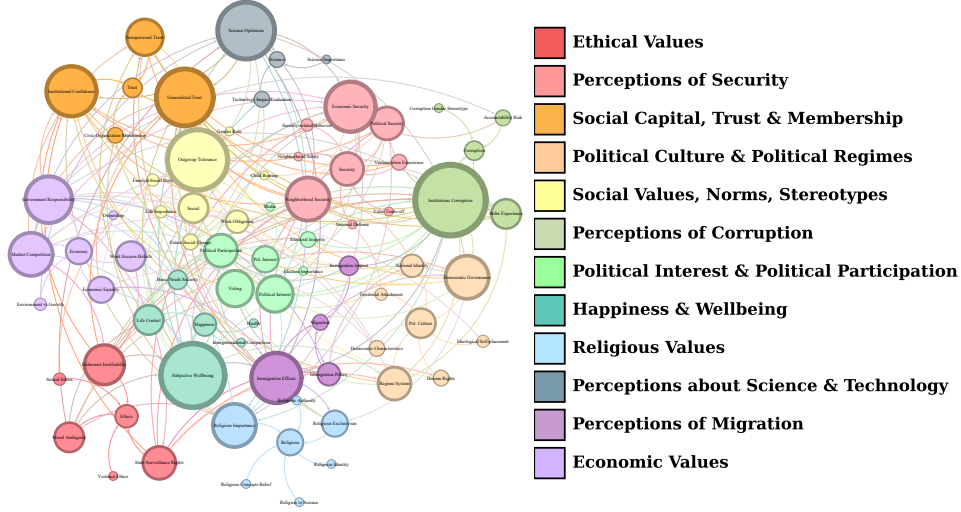


Figure 17. Final ontology structure with 76 classes and 150 object-property pairs. Node colors show the 12 top-classes, and node size scales with the sum of in-degree and out-degree, so that larger nodes mark sub-classes that are frequently instantiated in ontology triples and maintain rich relational connections to many other classes.

I. Ablation Study Details

I.1. VARYING THE NUMBER OF RETRIEVED INDIVIDUALS Full Figures

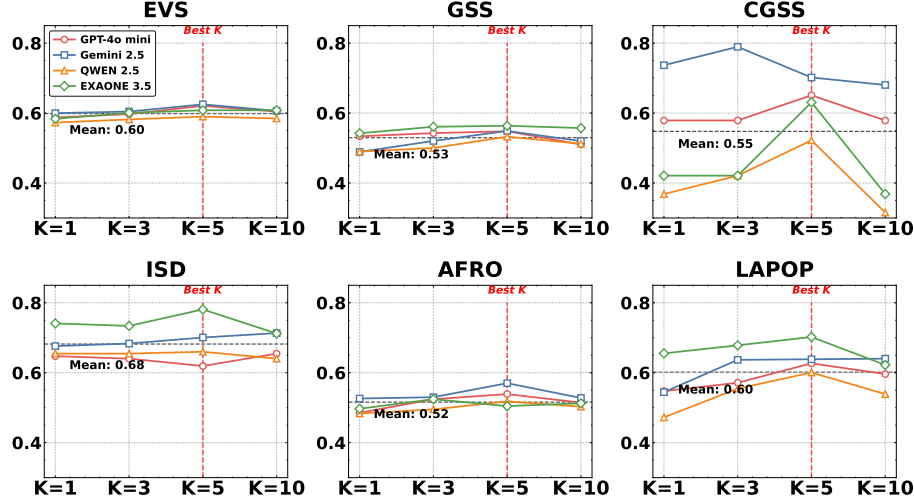


Figure 18. Detailed ablation study on retrieval size K across six regional datasets. Each subplot shows the performance comparison of four models (GPT-4o mini, Gemini 2.5, QWEN 2.5, EXAONE 3.5) across $K \in \{1, 3, 5, 10\}$. Red vertical dashed lines indicate the best K for each dataset, and black horizontal dashed lines show the dataset-specific mean accuracy. The results demonstrate that $K=5$ achieves optimal or near-optimal performance across most datasets, while $K=10$ often leads to performance degradation due to increased noise in the retrieved context.

I.2. IMPACT OF MULTI-PERSONA REASONING Full Table

Table 27. Detailed breakdown of accuracy scores by region for the full OG-MAR framework compared to the Single-Judge variant (referenced in Section 5.2.3). The highest score between the two methods for each region is highlighted in bold.

Model	Method	EVS	GSS	CGSS	ISD	AFRO	LAPOP	Avg. Acc.
GPT-4o mini	OG-MAR	0.6206	0.5480	0.6509	0.6192	0.5389	0.6268	0.6007
	Single-Judge	0.5773	0.6000	0.6440	0.6996	0.5293	0.5419	0.5987
Gemini 2.5	OG-MAR	0.6249	0.5489	0.7017	0.7007	0.5701	0.6385	0.6308
	Single-Judge	0.5870	0.6222	0.5960	0.6551	0.5411	0.6116	0.6022
QWEN 2.5	OG-MAR	0.5898	0.5325	0.5220	0.6599	0.5180	0.6005	0.5705
	Single-Judge	0.5266	0.5777	0.4067	0.6485	0.4494	0.5779	0.5311
EXAONE 3.5	OG-MAR	0.6080	0.5636	0.6307	0.7810	0.5045	0.7022	0.6317
	Single-Judge	0.5013	0.6444	0.4237	0.6900	0.4725	0.6444	0.5627

I.3. Additional Ablation Study: Impact of Retrieved Ontology Triples

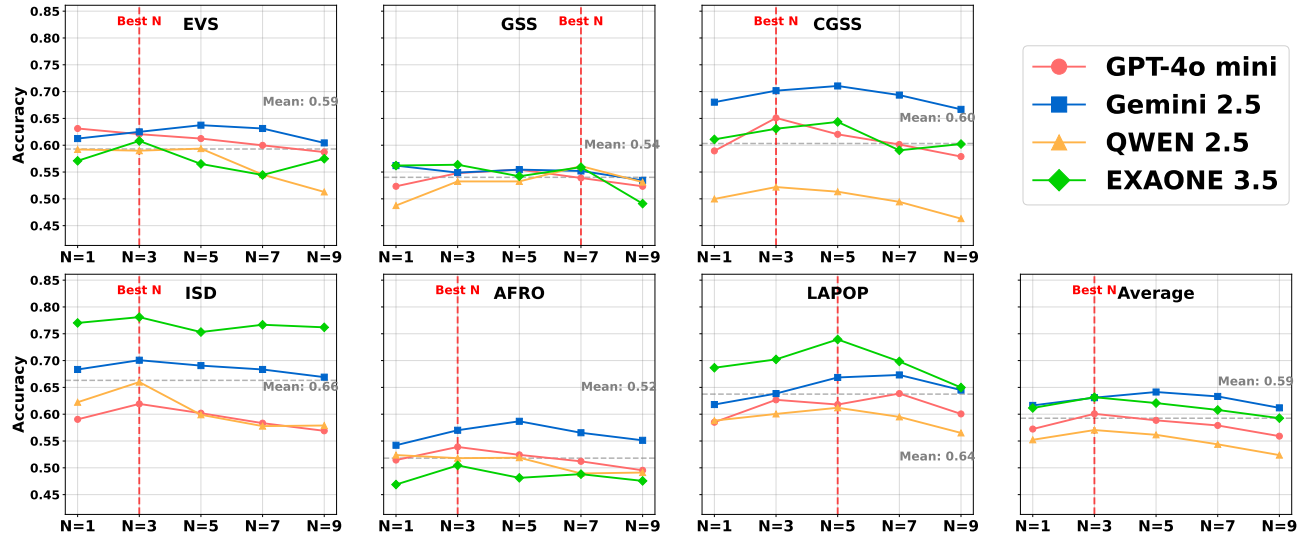


Figure 19. Ablation study on ontology triples retrieval size. Performance comparison across $N \in \{1, 3, 5, 7, 9\}$ for four LLM backbones on six regional datasets and their average. Red dashed vertical lines mark the Best N where average accuracy across all models peaks for each dataset. Gray dashed horizontal lines show the overall mean accuracy with values displayed. Results demonstrate that $N = 3$ achieves competitive or near-optimal performance across most datasets.

We conduct an additional ablation to examine how the number of retrieved ontology triples N affects **OG-MAR**’s performance across different regions and backbones. We vary $N \in \{1, 3, 5, 7, 9\}$ while keeping all other components of **OG-MAR** fixed, and report accuracy on six regional datasets and their average for GPT-4o mini, Gemini 2.5 Flash Lite, QWEN 2.5, and EXAONE 3.5. Figure 19 visualizes per-dataset trends together with the best-performing N and the overall mean accuracy.

Across datasets, $N = 3$ provides consistently strong performance and is either the best or very close to the best choice for most model–dataset pairs. Larger N sometimes brings small gains on specific datasets but often leads to plateauing or slight degradation, suggesting that adding too many ontology triples introduces noise rather than useful structure. Extremely small N (e.g., $N = 1$) tends to underperform, indicating that a minimal amount of ontology context is necessary for stable value reasoning. Taken together, these results support our default choice of $N = 3$ as a robust trade-off between leveraging ontology structure and avoiding over-retrieval noise.

I.4. Additional Ablation Study: Component-Level Analysis

To identify which components of **OG-MAR** drive its performance gains, we conduct a component-level ablation with three additional variants. **Without Ontology** removes all ontology context from the pipeline. In this setting, persona agents still have access to demographic information, but they reason from broad value summaries for the three predicted categories rather than from fine-grained ontology triples and individual value profiles. This condition directly answers the question of whether any structured ontology context outperforms having none, complementing the ontology triple ablation in Appendix I.3. **Without Demographic Retrieval (Single-Agent)** removes demographic retrieval entirely. The model has no access to the $K=5$ most similar respondents or their fine-grained value

profiles, which makes the persona mechanism uninformative. A single agent therefore reasons from ontology context alone. **Top-1 Demographic Retrieval (Single-Agent)** retrieves only the single most similar respondent and uses a single agent for reasoning. This variant introduces minimal demographic grounding under the same single-agent structure and provides a controlled comparison against **Without Demographic Retrieval (Single-Agent)**.

We note that **Without Demographic Retrieval (Single-Agent)** is distinct from the **Single-Judge** variant evaluated in Section 5.1.3. Single-Judge receives the same full input as **OG-MAR**, including the demographics and value profiles of $K=5$ similar respondents along with ontology context, and differs only in that it bypasses the persona-simulation stage to produce a final answer with a single judgment model. **Without Demographic Retrieval (Single-Agent)**, by contrast, eliminates demographic retrieval from the input itself, so only one agent operates on ontology context alone. The two variants therefore target distinct questions. Single-Judge asks whether multi-persona simulation is necessary given full input. **Without Demographic Retrieval (Single-Agent)** asks whether demographic information is necessary.

Table 28. Component-level ablation results. Average accuracy across six regional datasets for three ablation variants and the full **OG-MAR** framework. **(w/o) Ontology** retains demographic grounding but replaces ontology triples with broad category summaries. **(w/o) Demographic Retrieval (Single-Agent)** retains ontology context but removes all demographic retrieval and uses a single agent. **Top-1 Demographic Retrieval (Single-Agent)** retrieves one respondent under the same single-agent structure. Bold denotes the best result per row.

Model	(w/o) Ontology	(w/o) Demographic Retrieval (Single-Agent)	Top-1 Demographic Retrieval (Single-Agent)	OG-MAR
GPT-4o mini	0.5852	0.5667	0.5953	0.6007
Gemini 2.5 Flash Lite	0.6083	0.5713	0.6102	0.6308
QWEN 2.5	0.5567	0.5107	0.5508	0.5705
EXAONE 3.5	0.5858	0.6139	0.6146	0.6317

Table 28 reports average accuracy across six regional datasets. The full **OG-MAR** framework achieves the best result for all four backbones, confirming that performance gains arise from the combination of components rather than from any single one. Removing ontology context (**Without Ontology**) consistently lowers performance relative to **OG-MAR** on GPT-4o mini (-0.015), Gemini 2.5 Flash Lite (-0.023), QWEN 2.5 (-0.014), and EXAONE 3.5 (-0.046). This confirms that structured ontology triples provide meaningful guidance beyond broad category-level summaries across all backbones, directly establishing the benefit of any ontology context over none.

Removing demographic retrieval entirely (**Without Demographic Retrieval (Single-Agent)**) causes the largest performance drop for GPT-4o mini (-0.034), Gemini 2.5 Flash Lite (-0.060), and QWEN 2.5 (-0.060), falling below even the **Without Ontology** condition for these three models. This indicates that respondent-specific value profiles are the most critical input component for these backbones. Recovering to **Top-1 Demographic Retrieval (Single-Agent)** consistently improves accuracy above **Without Demographic Retrieval (Single-Agent)** for all four backbones, but does not fully close the gap to the full model on any of them, showing that limited demographic grounding partially but not fully substitutes for the richer multi-persona setup. On EXAONE 3.5, the pattern differs from the other three models. **Without Demographic Retrieval (Single-Agent)** (0.6139) outperforms **Without Ontology** (0.5858), indicating that this backbone relies more heavily on ontology structure than on demographic diversity. The gap between **Without Demographic Retrieval (Single-Agent)** and **Top-1 Demographic Retrieval (Single-Agent)** is negligible at 0.0007 for EXAONE 3.5, while both remain clearly below **OG-MAR** (0.6317), confirming that the full combination of ontology structure and multi-persona demographic grounding is necessary to achieve peak performance on this backbone.